

**ENSHAN CHEN**  
SELECTED WORKS

DENSITY  
LINKAGE  
DIVERSITY  
VITALITY

MASTER OF ARCHITECTURE  
TONGJI UNIVERSITY

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## URBAN

More problem solving orientated

Cities are all complex systems, containing numerous elements and relationships, so there are all constraints in it. I see urban design as an opportunity to solve problems like fragmentation, isolation, using strategies like linking, placemaking, thereby adding value to the built environment.

## ARCHITECTURE

More vision orientated

When it comes to a single building, the outside influence sometime diminishes to a minimum level. Then here designers' imagination matter. A creative vision can energize the building, turning the meaningless voids into space.

## RESEARCH

Problem identification & solution finding

Through researching we can deeply understand the problems underlying, and we can also use research to better our design. Rational research methodology is a counterpart to the emotional design methodology.

## CONTENT

### RESEARCH WORKS

#### DENSITY & MIXED-USE

Urban form of station areas

#### VISSIBILITY & ACCESSIBILITY

"Street eye" of Utrecht central station area

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Utrecht central station area analysis

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District Renewal

#### LINKAGE & PLACEMAKING

Railway Station Area Urban Design

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Office complex

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#### BUBBLES & RINGS

Kindergarten

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Shopping Mall

Residential district

Template Preschool Shop

Craftmanship analysis



# DENSITY & MIXED-USE

The urban form of railway station areas: an analysis based on density and function indexes

**Abstract:** Based on two typical density and function analysis frameworks, the data of 38 stations around the world are captured from the CadMapper website, the required indicators are extracted through the grasshopper software platform, and relevant diagrams are drawn and statistical analysis is conducted. The results show that there are obvious regional differences in the density of station areas around the world, and the degree of mixing is improved due to the improvement of residential and other functions.

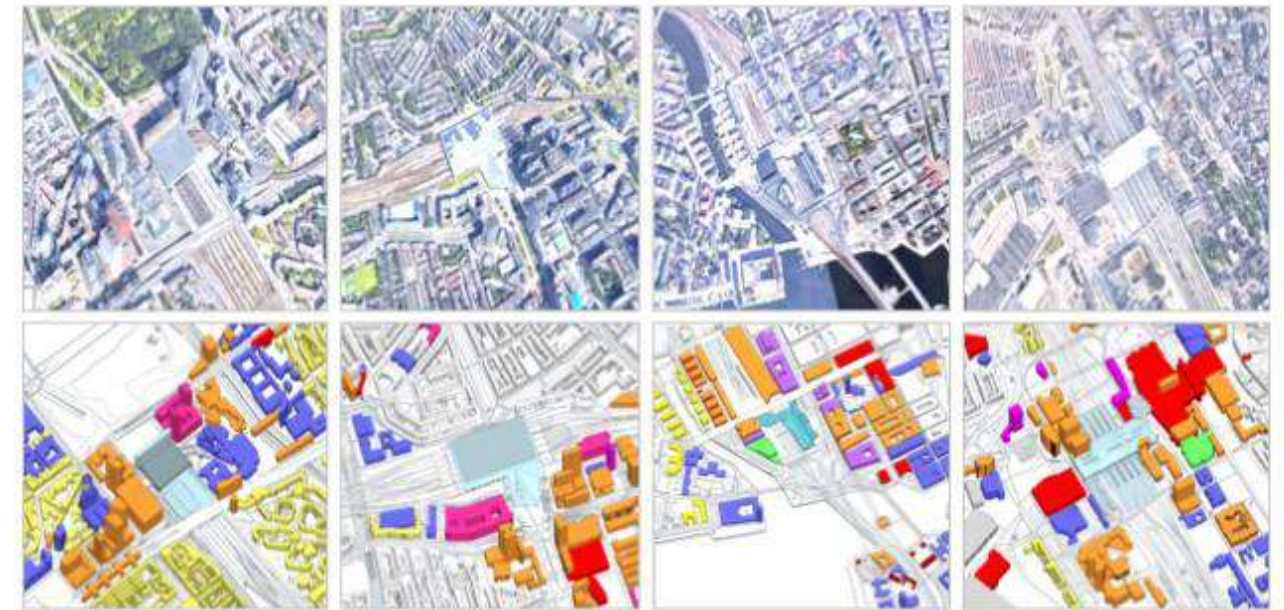


Figure 5, 6 Color the buildings based on Google Earth

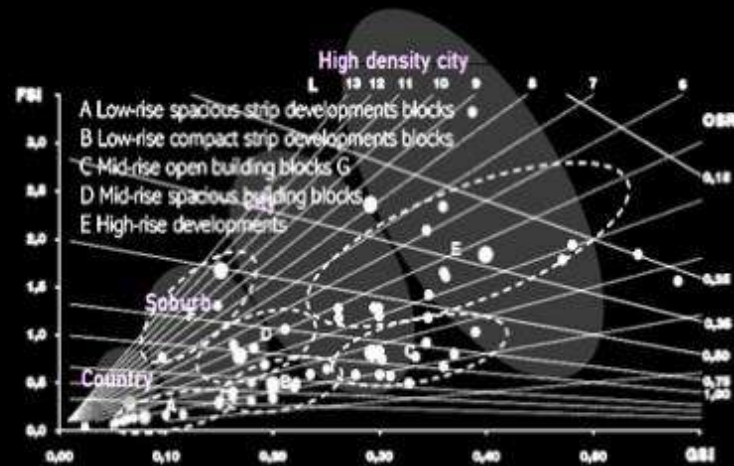


Figure.1. "spacemate"

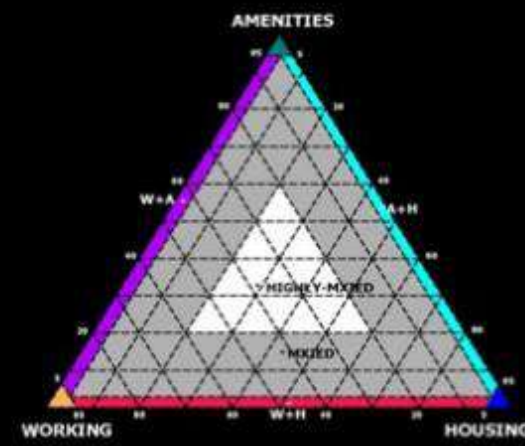


Figure.2. MXI



Figure.3. models downloaded from CadMapper

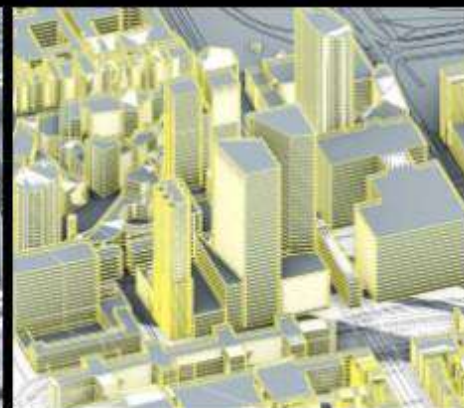


Figure.4. Floor slabs created by grasshopper

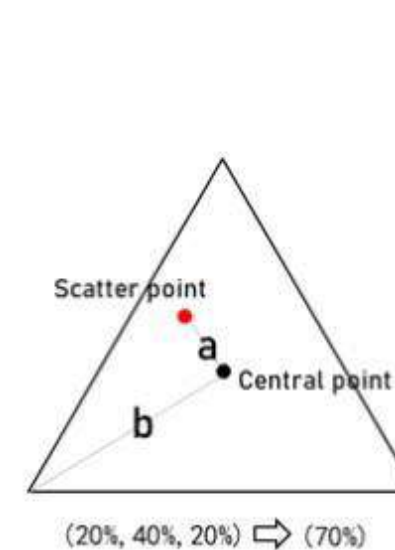


Figure.7 The index M

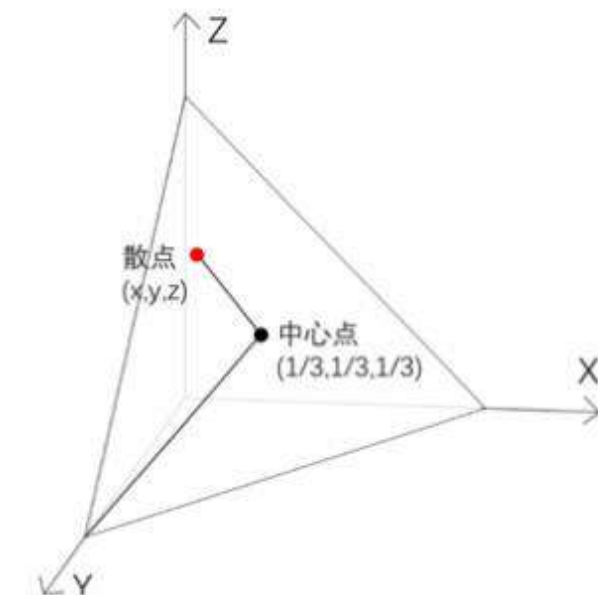


Figure 8 The mathematics of M value

## Defining a new index M to combine the separate MXI index

The original MXI is composed of three values, to make these values more understandable, we propose the index "M". Suppose the distance from the center point to a scatter point of the MXI diagram is "a", the distance from the center point to the end point is "b"(Figure 7), and the "M" index is defined as:

$$M=1-a/b$$

Assuming the MXI value of a station is (x, y, z), it can be derived:

$$M=1-3 \times \sqrt{(x-1/3)^2 + (y-1/3)^2 + (z-1/3)^2} / \sqrt{6}$$



| Station Areas                  | GSI   | FSI    | L      | OSR   | Facilities | Residence | Office | M     |
|--------------------------------|-------|--------|--------|-------|------------|-----------|--------|-------|
| Antwerpen                      | 0.546 | 5.085  | 9.314  | 0.089 | 0.545      | 0.032     | 0.423  | 0.536 |
| Berlin Zoologischer Garten sta | 0.388 | 3.299  | 8.504  | 0.185 | 0.569      | 0.035     | 0.396  | 0.527 |
| budapest western station       | 0.465 | 3.348  | 7.196  | 0.160 | 0.811      | 0.000     | 0.189  | 0.266 |
| chicago_millennium             | 0.375 | 10.875 | 29.007 | 0.057 | 0.457      | 0.002     | 0.542  | 0.497 |
| chicago_union station          | 0.561 | 11.265 | 20.078 | 0.039 | 0.388      | 0.055     | 0.557  | 0.558 |
| delft station                  | 0.409 | 3.824  | 9.357  | 0.155 | 0.719      | 0.154     | 0.127  | 0.421 |
| Edinburgh Waverley Station     | 0.494 | 4.131  | 8.361  | 0.122 | 0.557      | 0.186     | 0.256  | 0.659 |
| hague central Station          | 0.376 | 4.265  | 11.341 | 0.146 | 0.624      | 0.322     | 0.054  | 0.506 |
| Hakata station Station         | 0.283 | 3.510  | 12.404 | 0.204 | 0.208      | 0.386     | 0.405  | 0.812 |
| janpan_Shibuya Station         | 0.406 | 5.695  | 14.027 | 0.104 | 0.208      | 0.386     | 0.405  | 0.812 |
| japan_Futako-tamagawa Stati    | 0.180 | 2.295  | 12.749 | 0.357 | 0.422      | 0.182     | 0.396  | 0.772 |
| japan_Kokura Station           | 0.284 | 3.240  | 11.427 | 0.221 | 0.539      | 0.046     | 0.415  | 0.556 |
| japan_osaka Station            | 0.611 | 8.294  | 13.566 | 0.047 | 0.592      | 0.090     | 0.319  | 0.565 |
| Lille Europe Station           | 0.354 | 3.694  | 10.431 | 0.175 | 0.532      | 0.196     | 0.272  | 0.695 |
| Lisbon_oriente station         | 0.264 | 2.577  | 9.777  | 0.286 | 0.573      | 0.136     | 0.291  | 0.615 |
| london bridge station          | 0.407 | 5.023  | 12.328 | 0.118 | 0.300      | 0.216     | 0.484  | 0.763 |
| London_blackfriars station     | 0.389 | 4.547  | 11.684 | 0.134 | 0.359      | 0.191     | 0.450  | 0.773 |
| london_kings' cross            | 0.390 | 4.088  | 10.483 | 0.149 | 0.457      | 0.002     | 0.542  | 0.497 |
| london_liverpool street        | 0.539 | 5.963  | 11.062 | 0.077 | 0.548      | 0.132     | 0.319  | 0.639 |
| longdon_Charing Cross          | 0.385 | 3.821  | 9.932  | 0.161 | 0.484      | 0.359     | 0.158  | 0.715 |
| Madrid_atocha station          | 0.507 | 4.074  | 8.042  | 0.121 | 0.501      | 0.419     | 0.080  | 0.613 |
| milano Porta Garibaldi         | 0.361 | 3.987  | 11.032 | 0.160 | 0.353      | 0.277     | 0.370  | 0.914 |
| new york_Grand Central Term    | 0.916 | 17.176 | 18.742 | 0.005 | 0.345      | 0.004     | 0.651  | 0.439 |
| new york_HUDSON YARDS          | 0.425 | 5.768  | 13.565 | 0.100 | 0.241      | 0.071     | 0.689  | 0.447 |
| new york_Pennsylvania Static   | 0.638 | 10.304 | 16.158 | 0.035 | 0.441      | 0.048     | 0.511  | 0.568 |
| paris leshalles                | 0.545 | 5.128  | 9.413  | 0.089 | 0.500      | 0.358     | 0.142  | 0.688 |
| paris Austerlitz               | 0.300 | 3.168  | 10.552 | 0.221 | 0.393      | 0.138     | 0.470  | 0.699 |
| paris montpanasse              | 0.484 | 4.660  | 9.626  | 0.111 | 0.390      | 0.186     | 0.425  | 0.776 |
| Rotterdam                      | 0.426 | 4.690  | 11.004 | 0.122 | 0.231      | 0.474     | 0.295  | 0.782 |
| san francisco Transbay Transit | 0.581 | 11.649 | 20.033 | 0.036 | 0.358      | 0.039     | 0.604  | 0.509 |
| Switzerland_Zurich             | 0.386 | 3.866  | 10.016 | 0.159 | 0.473      | 0.228     | 0.299  | 0.781 |
| Switzerland Zukunft Bahnhof    | 0.413 | 3.707  | 8.971  | 0.158 | 0.452      | 0.279     | 0.268  | 0.821 |
| tokyo station                  | 0.561 | 10.649 | 18.993 | 0.041 | 0.132      | 0.203     | 0.666  | 0.498 |
| Shibuya Station                | 0.409 | 6.391  | 15.641 | 0.093 | 0.167      | 0.157     | 0.676  | 0.486 |
| Toronto Union Station          | 0.519 | 9.490  | 18.272 | 0.051 | 0.356      | 0.060     | 0.584  | 0.545 |
| utrecht central Station        | 0.395 | 4.365  | 11.043 | 0.139 | 0.369      | 0.283     | 0.348  | 0.922 |
| vallingby Station              | 0.182 | 2.562  | 14.052 | 0.319 | 0.352      | 0.627     | 0.021  | 0.474 |
| washington DC union station    | 0.318 | 3.829  | 12.039 | 0.178 | 0.815      | 0.120     | 0.065  | 0.276 |

Table.1 Density and function data

|            |         | GSI      | FSI      | L        | OSR       | Facilities | Residence | Office   | M        |
|------------|---------|----------|----------|----------|-----------|------------|-----------|----------|----------|
| GSI        | Pearson | 1        | .799**   | 0.278548 | -.860**   | -0.06254   | -.372*    | .357*    | -0.22971 |
| FSI        | Pearson | .799**   | 1        | .783**   | -.787**   | -0.3023    | -.424**   | .607**   | -0.29627 |
| L          | Pearson | 0.278548 | .783**   | 1        | -.468**   | -.403*     | -0.29086  | .586**   | -0.24732 |
| OSR        | Pearson | -.860**  | -.787**  | -.468**  | 1         | 0.204197   | .358*     | -.469**  | 0.178624 |
| Facilities | Pearson | -0.06254 | -0.3023  | -.403*   | 0.2041966 | 1          | -0.293723 | -.630**  | -.428**  |
| Residence  | Pearson | -.372*   | -.424**  | -0.29086 | .358*     | -0.29372   | 1         | -.557**  | .483**   |
| Office     | Pearson | .357*    | .607**   | .586**   | -.469**   | -.630**    | -.557**   | 1        | -0.02071 |
| M          | Pearson | -0.22971 | -0.29627 | -0.24732 | 0.1786243 | -.428**    | .483**    | -0.02071 | 1        |

\*\* . At level 0.01, the correlation was significant.  
\* . At level 0.05, the correlation was significant.

\*\* . At level 0.01, the correlation was significant.  
\* . At level 0.05, the correlation was significant.

Table.2. Correlation analysis of each index

### General pattern of density

There are strong regional differences in density distribution. The Americas station areas have the highest GSI, FSI, and L; Japan, which people think is very crowded conventionally, has station density indicators distributed at all levels; Europe, where traditional protection is highly concerned, station areas have medium to low GSI and FSI values, and the value of L is low.

Overall, the more building functions in the station area are disproportioned towards work and facilities, and the proportion of residential functions is relatively low.

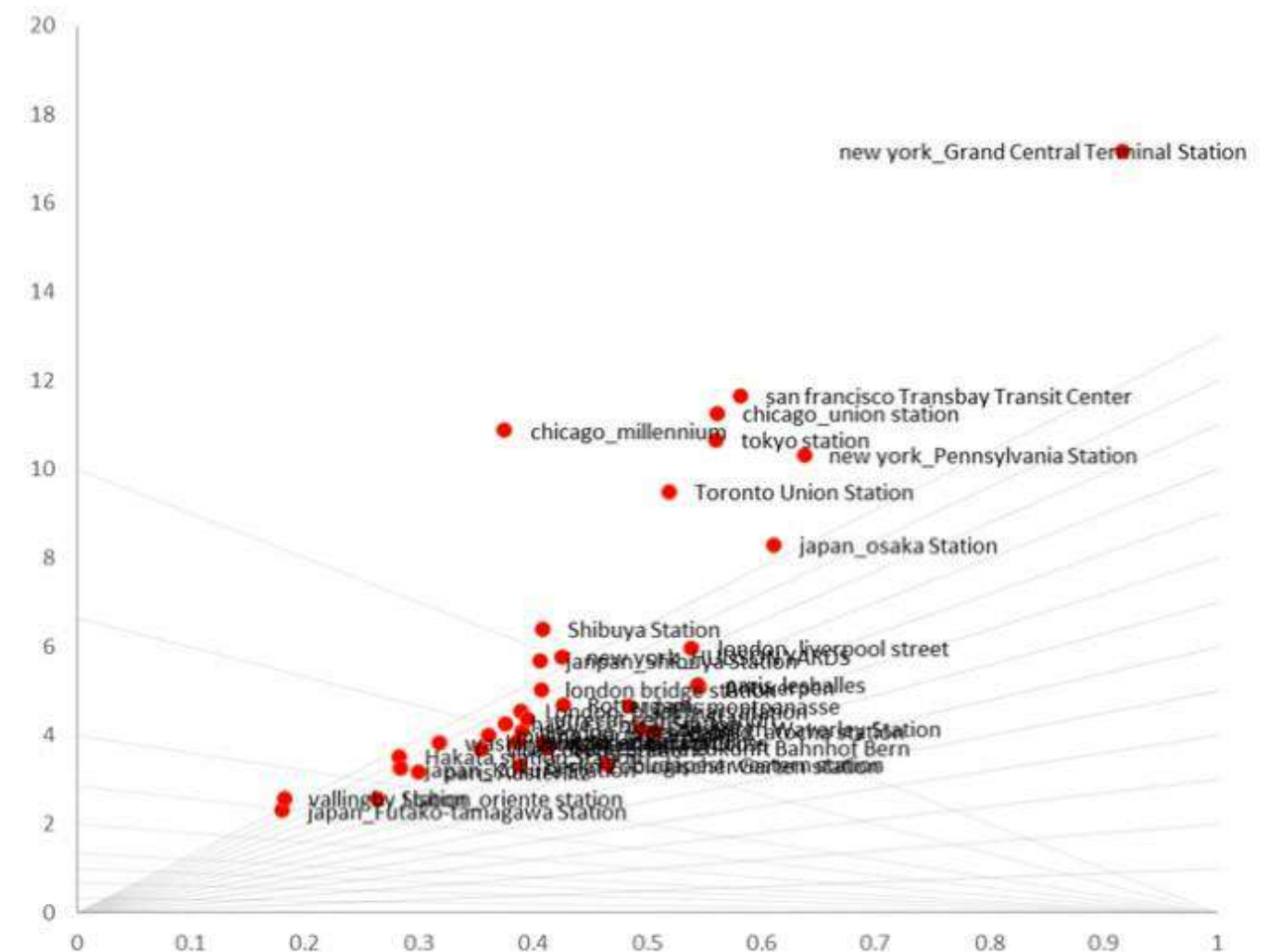


Figure.5: The spacemate diagram based on the data collected

### Case examples of density

In specific cases, though as the station with the largest passenger flow in the world, Shinjuku Station has all density indicators at an intermediate level; the GSI of New York Grand Central Station reaches 0.92, the L is 17.2, thereby turning this station area to be the most densified one among all the cases. (Picture 9, 10)



Figure.9. Shinjuku Station

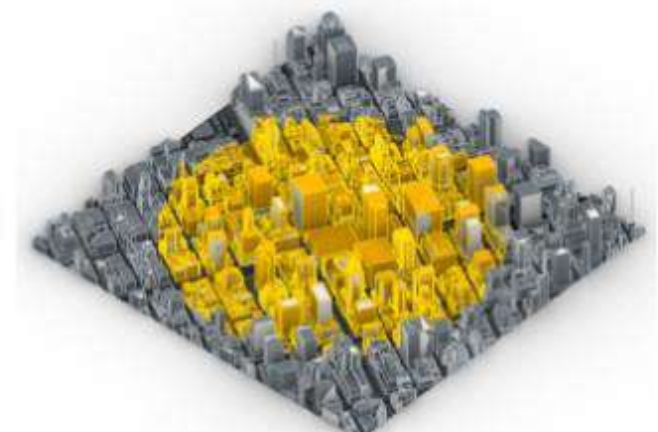


Figure.10. New York Grand Central Station



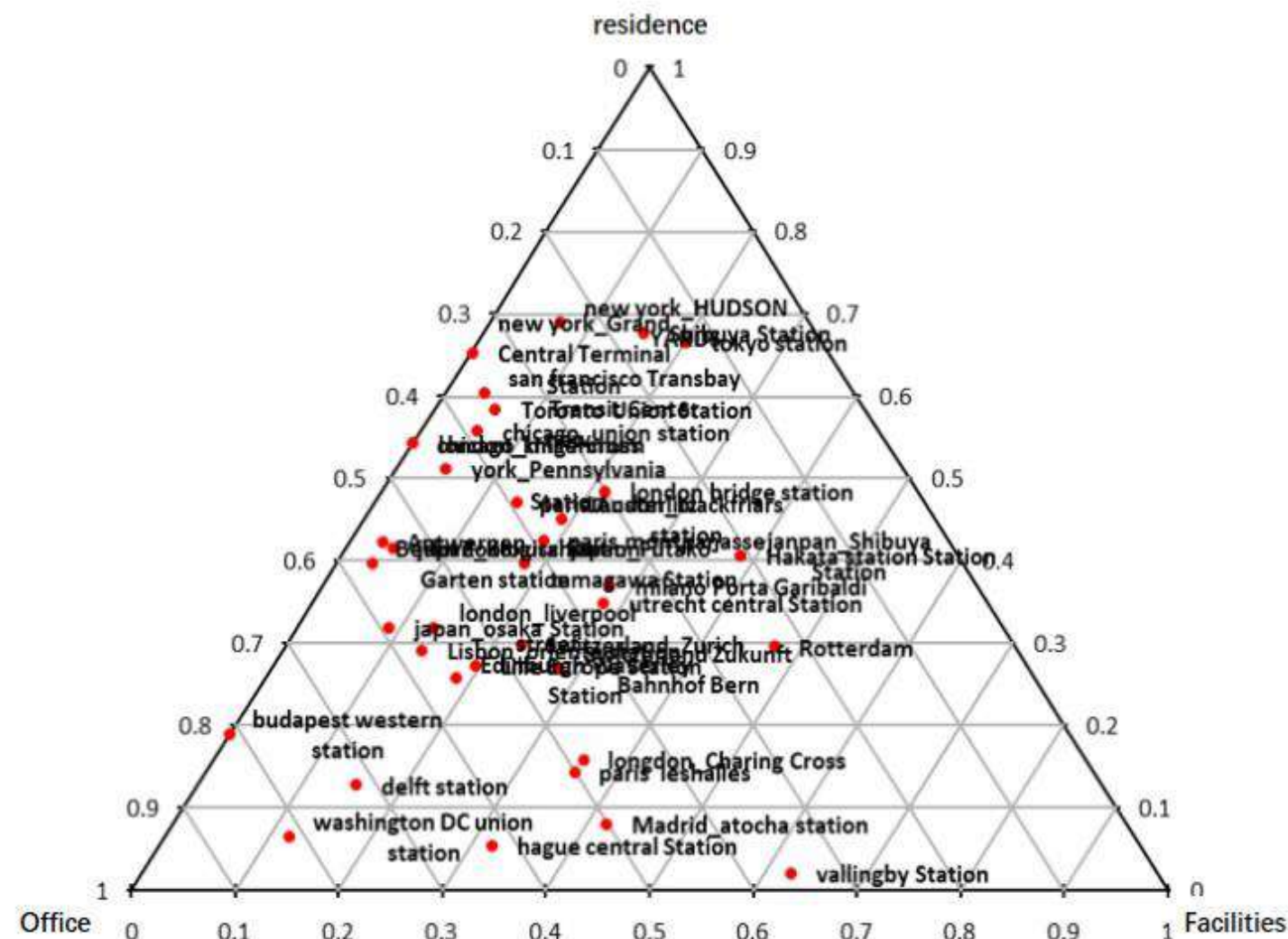


Figure.6. The MXI graph based on the data collected

### General pattern and cases of function

The American cases contain a high proportion of office functions, which is consistent with the phenomenon that American trains are mostly used for short-distance office commuting. Europe is represented by countries such as the Netherlands, with a high degree of function mix, and stations both as transportation nodes and as places of activities are strongly reflected (Figures 10 and 11).



Figure.11. The east side of the utrecht central station area



Figure.12. The west side of the utrecht central station area

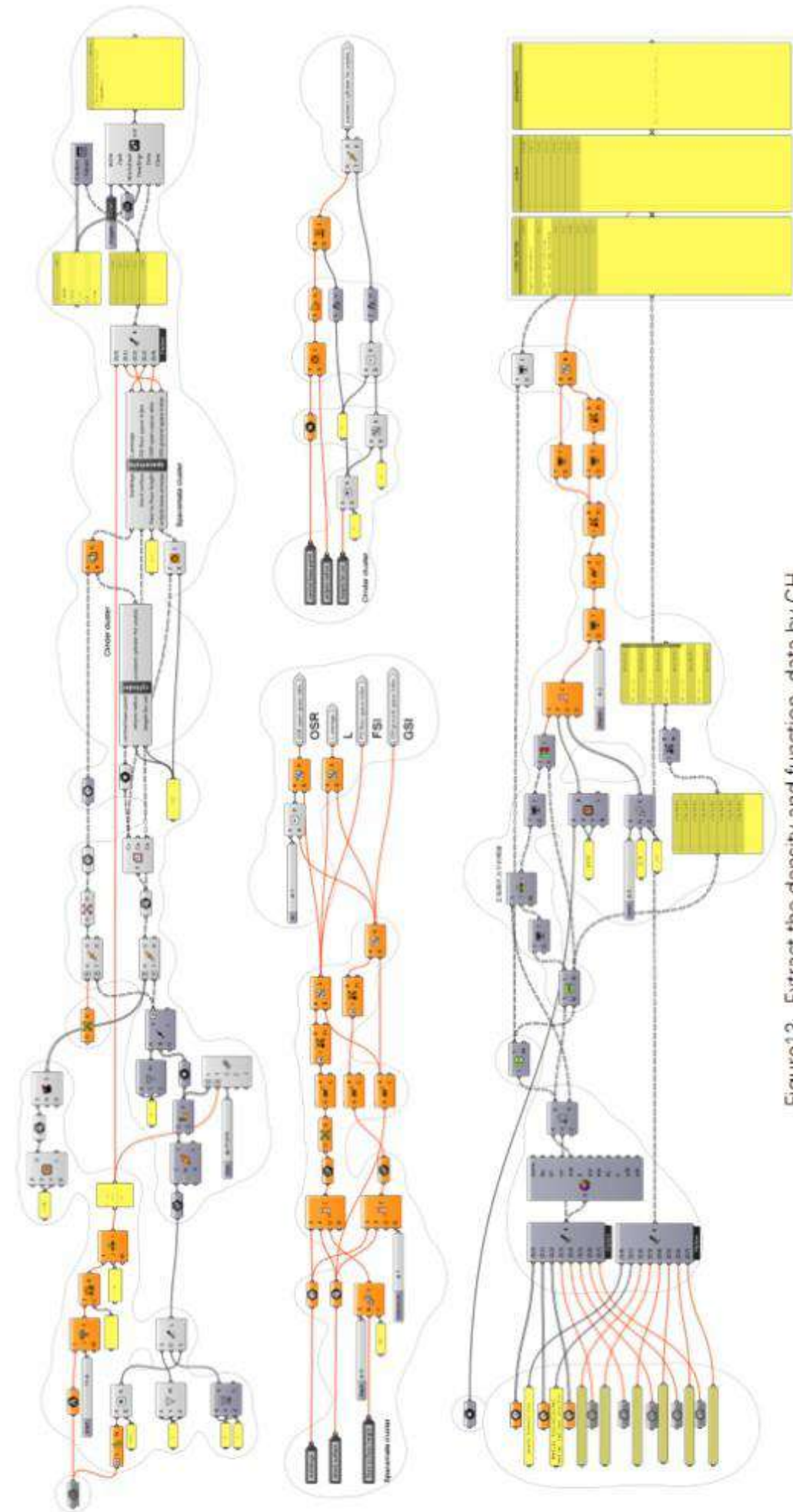


Figure13. Extract the density and function data by GH



# VISIBILITY & ACCESSIBILITY

## Analysis of Urban Security Based On Visibility: A Case of Utrecht Railway Station Area

**Abstract:** Urban open spaces, become more secure under the visual supervision of the surrounding environment. This article view analysis to put forward the concept of Visual Monitoring Rate to describe visual supervision. Taking the Utrecht railway station area as an example, it quantitatively analyzes the space security and observes the dynamic relationship between visibility and space security, verifying the validity of the whole approach.

### Isovist 2D and Isovist 3D

The concept of isovist (field of view) from the discipline of geography is currently used in architecture and urban design field for view analysis. But for space analysis, more factors need to be considered.

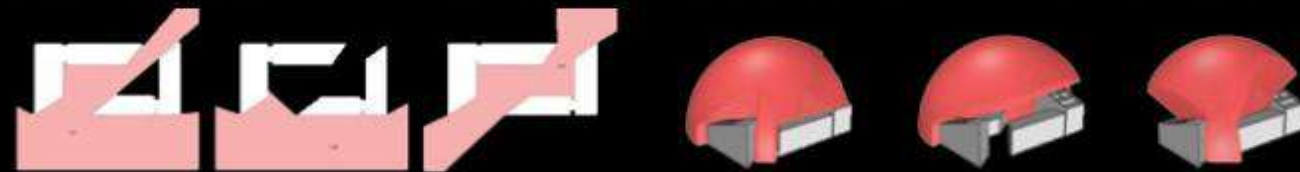
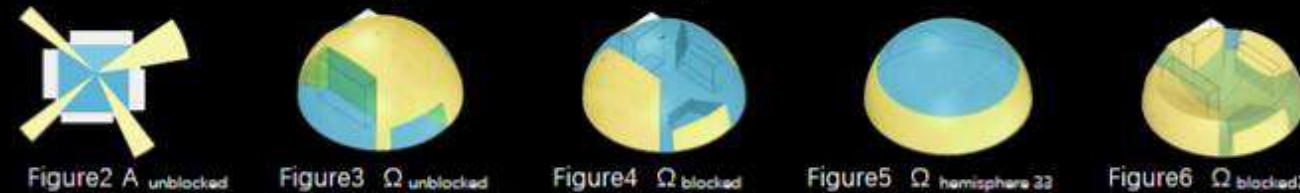


Figure 1 Isovist 2D and Isovist 3D

### Defining the visibility

Since the line of sight is reciprocal, when one ray of sight from the origin endpoint a is blocked by the building, it also means that we can see point a from the building. Considering the characteristics of the human vision, we set the downward viewing angle range as 0–30°, that is, the comfortable angle of the person near the building window is no more than 30° when looking down. Besides, the recognition ability of the human eye has a distance limit; here we set the distance as 30 meters.

$$V_{\text{monitoring } 33} = \Omega_{\text{blocked } 33} / \Omega_{\text{hemisphere } 33} \times 100\% \text{ (Equation 4)}$$



### Ladybug software platform

Divide the ground into 4mx4m squares, and calculate the visibility value of every square center point. Then, we get a map of visibility.

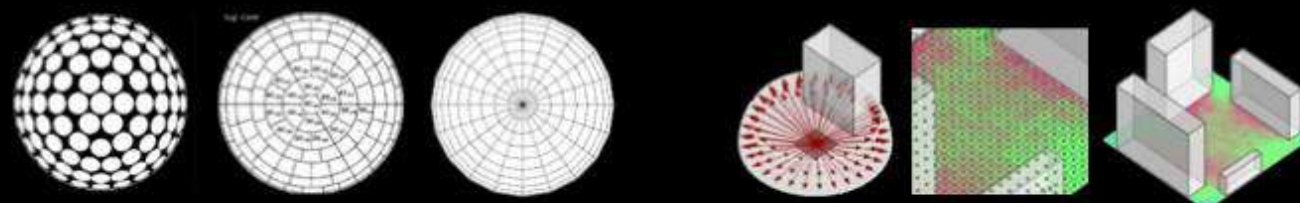


Figure 8 Tregenza sky subdivision

Figure9 The principle of view analyzing function in Ladybug software

### Modeling the Utrecht railway station area

This research obtains spatial information during various periods from the following sources: Utrecht Archives (online), Google Earth, CadMapper map information website, CU2030 project official website. After the information is compared, three representative time points of 1968, 1992, and 2020 are selected. With the plan outline and height information of each building, we built a three-dimensional urban model of this area (Figure 8)



Figure8. Utrecht Central Station area in different times

### Calculating the visibility map

From the results of the analysis, we can see that after the urban renewal in the 1960s and 1970s, along with the decline in building coverage, the visibility of public spaces has also decreased significantly. In different periods, the total Visual Monitoring Rate in the area averaged differently: 1961: 82.37%, 1992: 76.67%, 2030: 80.07%.

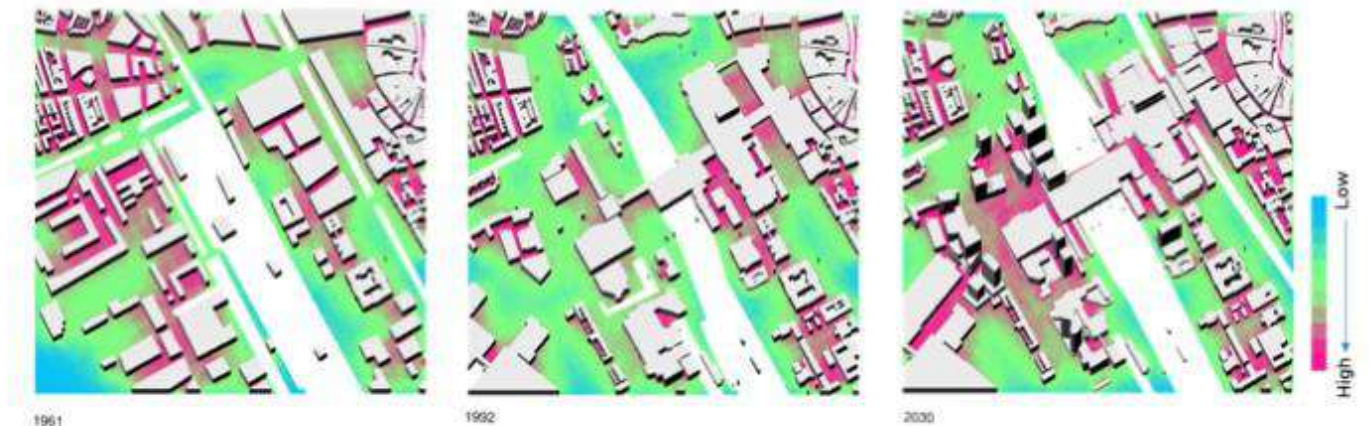


Figure10 Visual Monitoring map of utrecht central station area in different times



### Space quality identification by machine – the axis in the Utrecht

In the second urban renewal plan formulated in 2003, an urban axis connecting the east and the west was planned (Figure 11). In the implementation of the 2030 version of the plan, the axis was not obvious to the naked eye in terms of architectural form or spatial form. However, through the map, we can clearly recognize that the quality of view on this axis is consistent and uniform, which is significantly higher than that of the surrounding area (Figure 12)

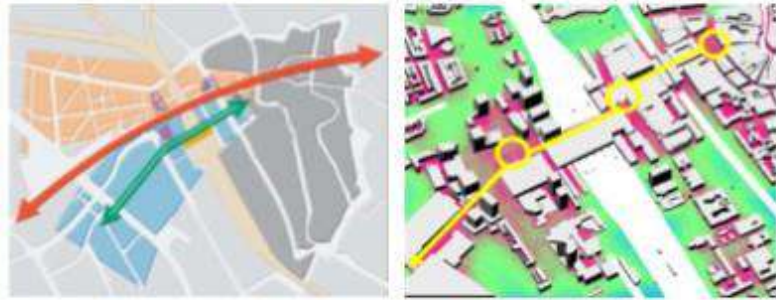
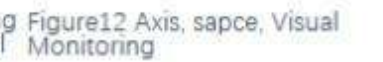
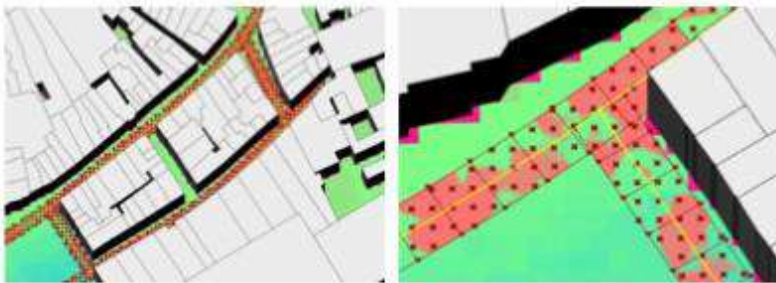


Figure 11 The City Axis and the Greening Axis in the second plan of city renewal



### Data standardization

Collect the data from 2D map to the street central lines, to draw the visibility map in the following figure13. so that we can overlap the visibility map with the accessibility map, and endows visibility the potential to overlay with other space quality analysis based on road or block network.



Collect the data from 2D map to the street central lines to draw the visibility map in the following figure13

### Overlap accessibility and visibility

Use "Spatial Design Network Analysis" software (sDNA) to calculate the accessibility, and converted the 2D visibility map into line map so that we can have the unified format of data for further analysis. After the two maps are overlapped, we can evaluate the quality of street space: which spaces have high accessibility but with no satisfactory visibility, and which spaces have high visibility, while the accessibility needs to be improved, etc

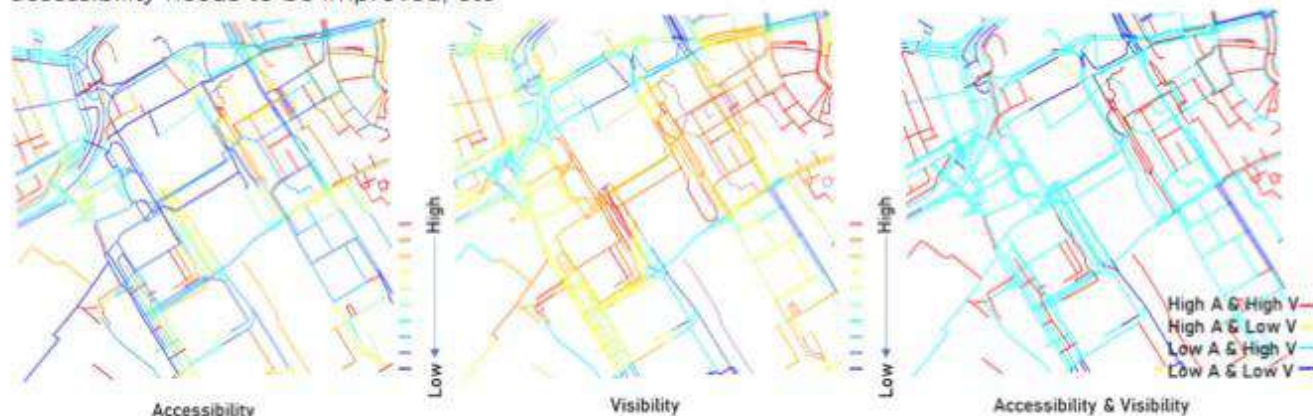


Figure13 Overlay analysis of Accessibility and Visibility

## SHAKEHOLDERS & ELEMENTS

— research on the station-city integration procedure of Utrecht central railway station

**Abstract:** Utrecht's station area renewal is a typical case in the Netherlands. Over the past three decades, participants have played a long game and successfully renewed the original deteriorating station area by reorganizing the city elements. By studying related Dutch official website materials and papers, this research demonstrates the driving forces behind the development and their impact on physical space.

### The history of the Utrecht railway station area development

| Years | Events                                                                                                     | Participants changes          | Relevant plans                                        | Background policies                                         |
|-------|------------------------------------------------------------------------------------------------------------|-------------------------------|-------------------------------------------------------|-------------------------------------------------------------|
| 1984  | Freight company moved                                                                                      |                               |                                                       |                                                             |
| 1986  | Hospital moved                                                                                             |                               | Klunder plan, kat&jong plan, DROarea list             |                                                             |
| 1987  | Facade Renewal plan Station building update                                                                |                               |                                                       |                                                             |
| 1988  | Tentative joint development statement                                                                      | the municipality, NS, ABP, JB |                                                       | VINO                                                        |
| 1989  | Atelier work studio                                                                                        |                               | "UCP Future Vision"                                   |                                                             |
| 1990  | Established the official UCP development platform                                                          |                               | NGO group BOCP is established                         | The second national transportation planning New Key Project |
| 1991  |                                                                                                            |                               |                                                       | VINEX                                                       |
| 1993  |                                                                                                            | ABP, JB withdrew              | 93 master plan                                        |                                                             |
| 1994  | Established UCP B.V.                                                                                       | Three major developers joined |                                                       |                                                             |
| 1995  |                                                                                                            |                               | Conceptual plan of space function                     |                                                             |
| 1996  | Established UCP administrative Platform and piggy bank model Riek joined                                   | ABP, JB rejoined              |                                                       |                                                             |
| 1997  | Government promised 2.4 billion subsidy "Livable Utrecht" party established                                |                               | "Temporary Urban Design" VSO "Final Urban Design" DSO |                                                             |
| 1998  | "Livable Utrecht" became one of the three major political parties                                          |                               |                                                       |                                                             |
| 1999  | Lobbying the state government                                                                              |                               |                                                       |                                                             |
| 2000  | UCP disintegrated "Restart plan" got cancelled later "Livable Utrecht" becomes the largest political party | JB, ABP withdraw              |                                                       |                                                             |
| 2002  | referendum, option A was selected The project restarts as CU2030                                           | ABP, JB rejoined              | Option A, Option 1                                    |                                                             |
| 2004  |                                                                                                            |                               | 04 masterplan                                         |                                                             |
| 2005  | UPL survey recommendations                                                                                 |                               |                                                       |                                                             |
| 2006  |                                                                                                            |                               | Structural Plan                                       |                                                             |
| 2014  | Selected as a smart sustainable district                                                                   |                               |                                                       |                                                             |
| 2018  | NS above track development intention                                                                       |                               |                                                       |                                                             |



## ▲ The history of the Utrecht railway station area development

There are many parties involved in the development of the Utrecht station area: the municipality, the NS (the Dutch national railway company), ABP Development Company (the owner of the Catharijne Shopping Center and surrounding plots 2), Jaarbeur Company (the owner of the Jaarbeur Convention and Exhibition Center and surrounding plots), the BOCP (a non-governmental organization), the state government, etc.

### The element interaction

In the development of urban design theories, different scholars have tried to summarize its elements. However, the various elements of a city are not parallel and isolated, but are intricately constrained.



Figure 4 the interaction of different city elements during the development

### Interests priorities

Residents and local governments pay more attention to regional housing and space quality, while developers are pursuing the commercial vitality and office rental and sales income. Railway companies focus on macro-level transportation efficiency. The central government focus on the overall urban and station synergy (Figure 4).

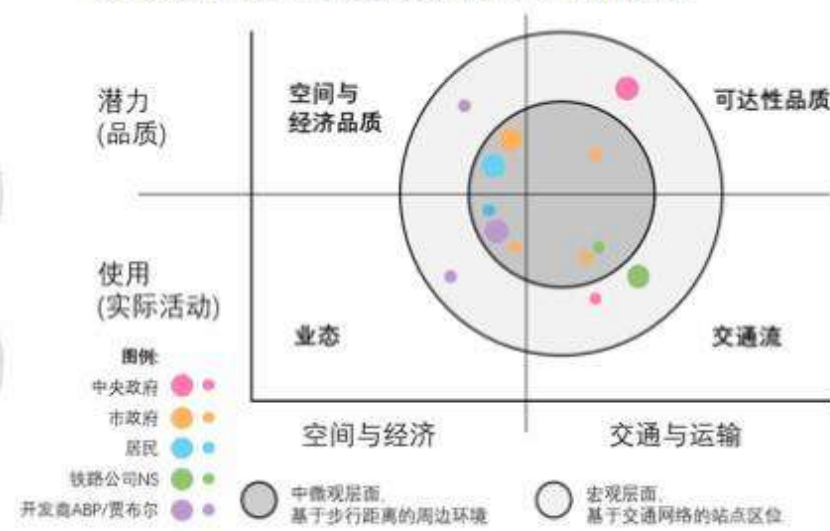


figure 5 different stakeholders and their main interest. illustrated based on the Dutch Nod-Place theory.

## The history line of the Utrecht railway station area development

Different stakeholders and users tried to pursue their own interests in the development progress, which can be reflected on different aspect of city elements.

| Stakeholder                                  | Main demands                                                       | Public space and architecture              | Transport                                                                                                       | Function                                                | Landscape                                   | History/culture                                |
|----------------------------------------------|--------------------------------------------------------------------|--------------------------------------------|-----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|---------------------------------------------|------------------------------------------------|
| <b>Residents</b>                             | Livable/Walk-friendly                                              | Update/Newly built community square/Mosque | Pedestrian bridge/Station city corridor<br>Urban corridor axis/transformation of motor vehicle lanes to tunnels | Housing                                                 | Park restorations<br>Boulevard<br>New canal | restoration of moat<br>building height control |
| <b>Passengers<br/>Commuters<br/>Tourists</b> | Convenient transportation/City recognizability/Consumer experience | Building integrity/Street wall control     | Station integration<br>Central Avenue Axis                                                                      | Hotel/Theater/Casino                                    |                                             | Artistic Creation                              |
| <b>Developers</b>                            | Commercial benefits                                                | Office plaza                               | Two-story platform                                                                                              | Office/commercial/above track development (in planning) | Residential building along the river        |                                                |
| <b>Government</b>                            | Security/city status                                               |                                            | "Railway Port"                                                                                                  | Mixed-use/Hospitality                                   |                                             |                                                |

## Common appeal: Improving safety, and the element organization

the Utrecht station area was updated in the 1960s and 1970s with space separated by traffic and functions isolated from each other. It soon became unsafe due to cars on the ground and insufficient pedestrian. So in 04 plan, the 11 sub-areas all equipped with residences for "street eye" monitoring, and 4 areas near the station are planned to set up shelters.



## Three pairs of main contradictions, and the element organization

### Contradiction 1: Residential qualities vs Development benefits

Solutions: the mixed-use, developing on the track, adding Community facilities, etc.



### Contradiction 2: Tourist Experiencing vs Commercial Value

Solutions: channel the obstructed passage, enhance City identification, etc



### Contradiction 3: Slow traffic vs Locomotive traffic

Solutions: restrict cars, promote public transit.







## U01 “EXERCISABLE” CITY

A residential districts design

Year: 2018

Location: Shanghai, China

Type: urban design

Category: Academic projects

Role: collaborative work, as the leader of 4 members

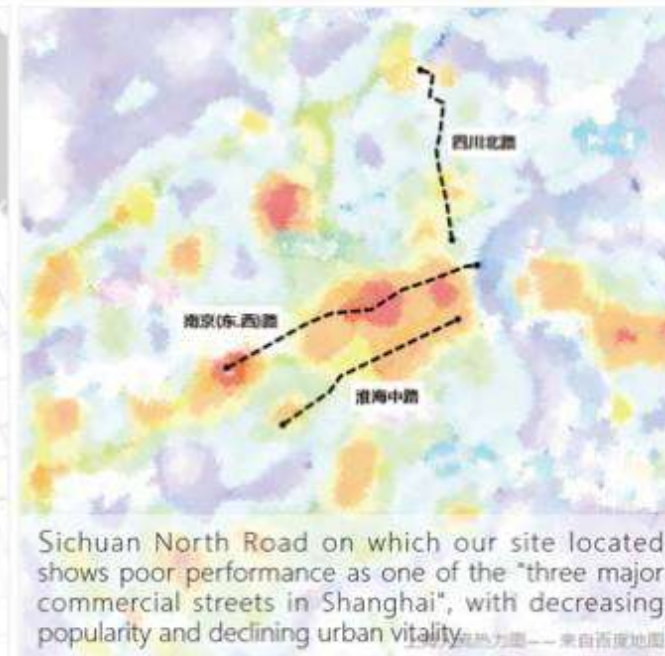
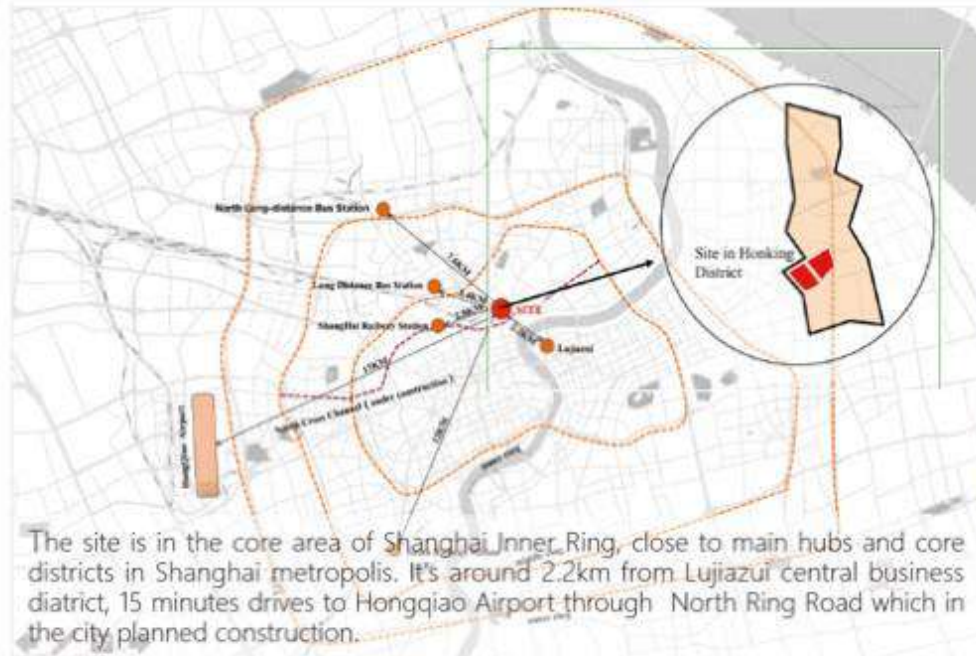
This is a district with historical building, subway station, and is adjacent to one historical street and one main traffic street. Concerning these elements, we start off from the area revitalization, setting this district as an “exercisable” city where every important spot is connected with the two loop, one on the ground and one in the sky.





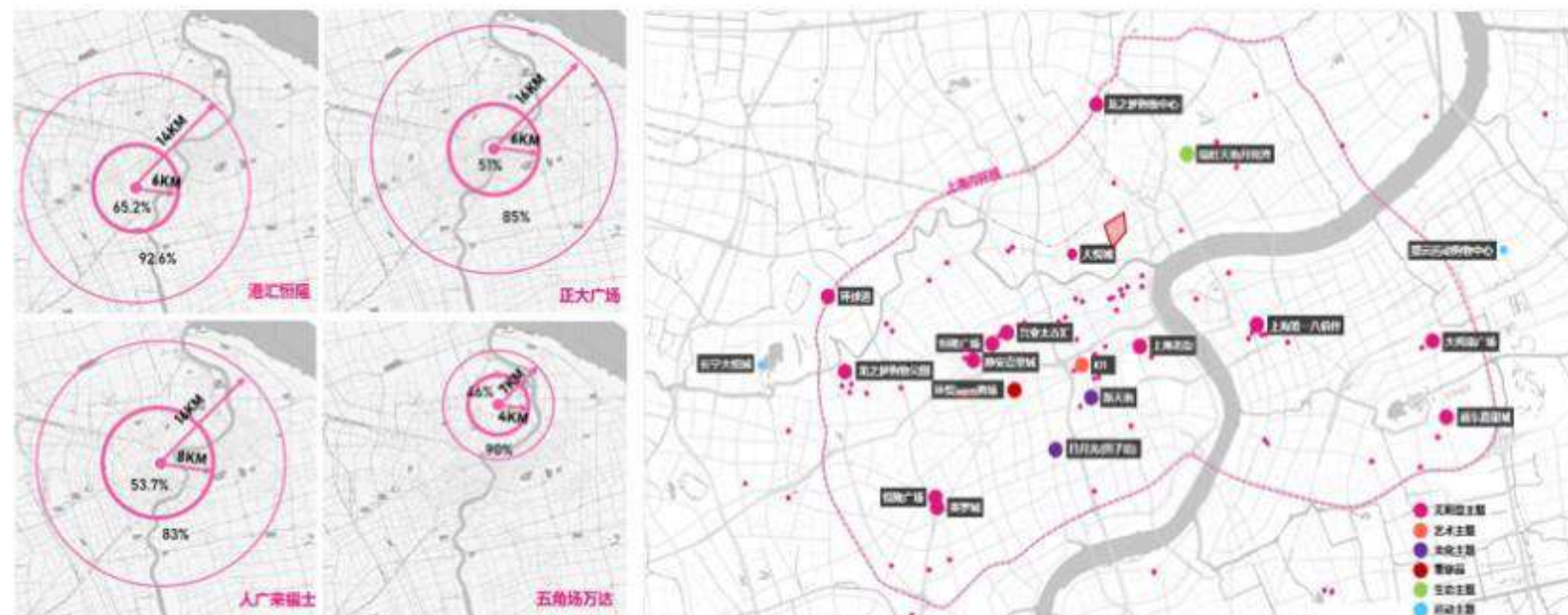
### Commercial decline

While in a key location at Shanghai, Sichuan North Road has declined as one of the previous three major commercial streets. The vitality here is far more less than the other two now.



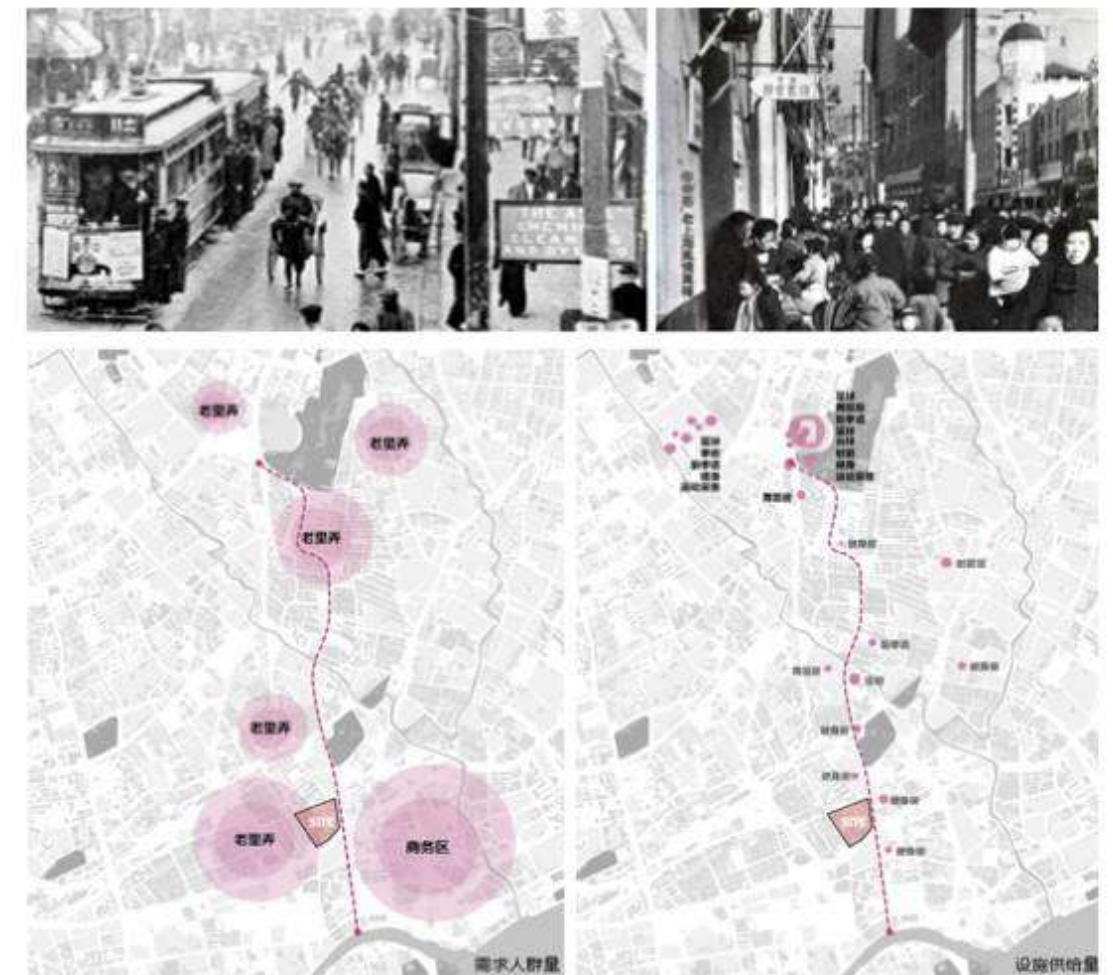
### A mall with sports theme

Potential components in the shopping mall commercial shows that sports theme would be a better choice.



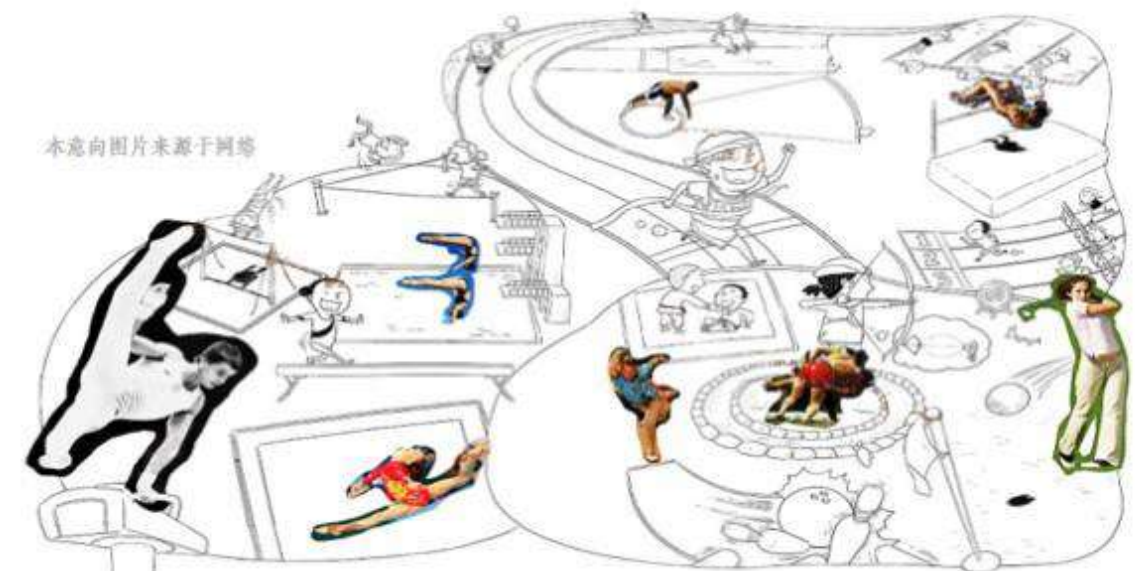
### Be a facility to the community

The site should be combined with surrounding facilities to support residence nearby to maximize the resources and bring vitality to the local area.



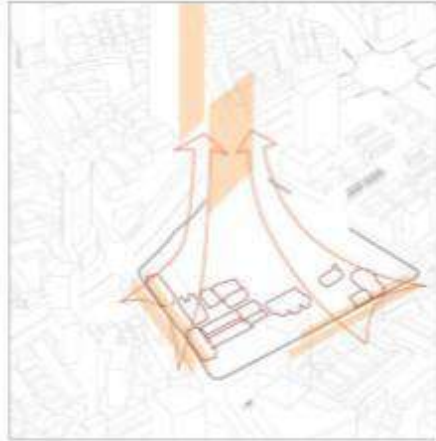
### The Sports Loop

Attract different age groups, with various social roles to this area, bring plenty of energy to this region. Improve and optimize people's life philosophy, promote healthy urban civilization.

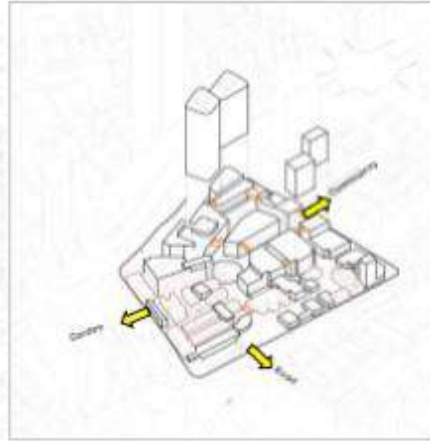




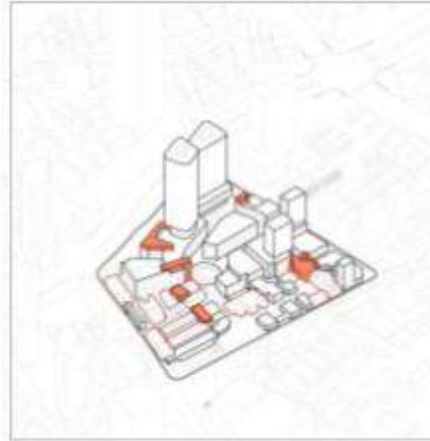
**Sept1:** Set heights regarding to the urban interface



**Sept2:** Link three roads and gardens to the site



**Sept3:** Insert several sports facilities in to the block



**Sept4:** Link all entrances and facilities with two main loops



### Placemaking

There are two main loops linking all the facilities together, we want to create small landmarks which are both identical and functional. The existing elements inspired the newly-built ones, by mirroring and duplicating.



1



2



3



4

通过人均商业面积测算的商业容量最大值, 设为A

项目商业面积=2.34m<sup>2</sup>/人×高规格消费人数

| 城市                        | 上海   | 无锡 | 青岛  | 宁波  | 深圳  |
|---------------------------|------|----|-----|-----|-----|
| 人均商业面积(m <sup>2</sup> /人) | 2.14 | 1  | 1.2 | 0.5 | 1.2 |

高规格消费人数为540K, 总人数为222万

通过人均可支配收入测算的商业容量最大值, 设为B

项目商业面积=居民的可支配收入总额÷10%÷4000万元/万m<sup>2</sup>

| 城市          | 上海   | 北京   | 天津   | 杭州   | 宁波   |
|-------------|------|------|------|------|------|
| 人均GDP(万元)   | 12.4 | 12.9 | 16.2 | 11.3 | 12.3 |
| 人均可支配收入(万元) | 6.2  | 6.7  | 8.1  | 5.6  | 6.2  |

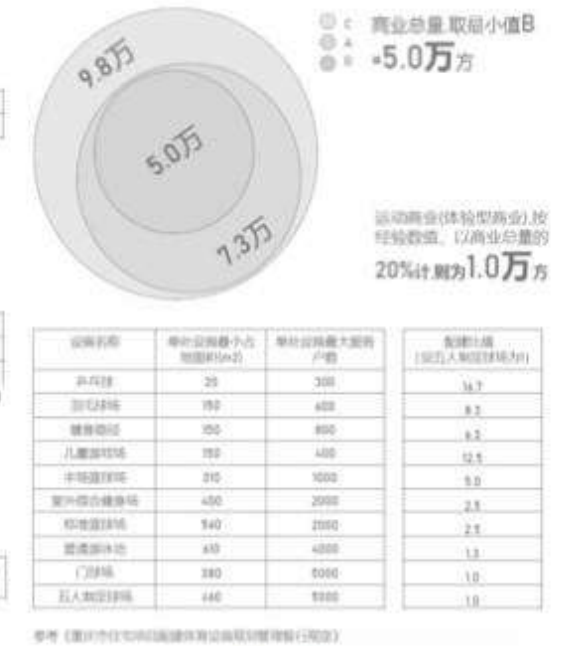
项目商业面积=222万×6.2÷10%÷4000=346.5万m<sup>2</sup>

通过地块面积测算的商业容量最大值, 设为C

项目商业面积=用地面积×建筑密度×(2-4层)

| 城市                     | 上海   | 北京   | 天津   | 杭州   | 宁波   |
|------------------------|------|------|------|------|------|
| 用地面积(万m <sup>2</sup> ) | 12.4 | 12.9 | 16.2 | 11.3 | 12.3 |
| 建筑密度                   | 0.3  | 0.3  | 0.3  | 0.3  | 0.3  |
| 商业建筑层数                 | 2-4  | 2-4  | 2-4  | 2-4  | 2-4  |

项目商业面积=12.4×0.3×3=1.116万m<sup>2</sup>



The commerce volume in this project is estimated and calculated in several ways to ensure that it will has sufficient customers.



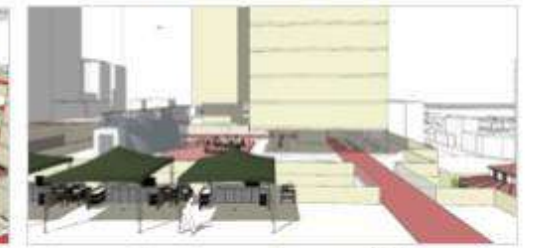
Courts on the roof 1



Lanes under roof 1



Courts on the roof 2



Lanes under roof 2



Typical road type designs



Underground spaces, Green spaces, Development volume, Logistics





This "Yihongyuan" building is an elegant and historical one, but currently is been surrounded with all the freely constructed buildings. We "save" it from this cram, and enhance its identification.



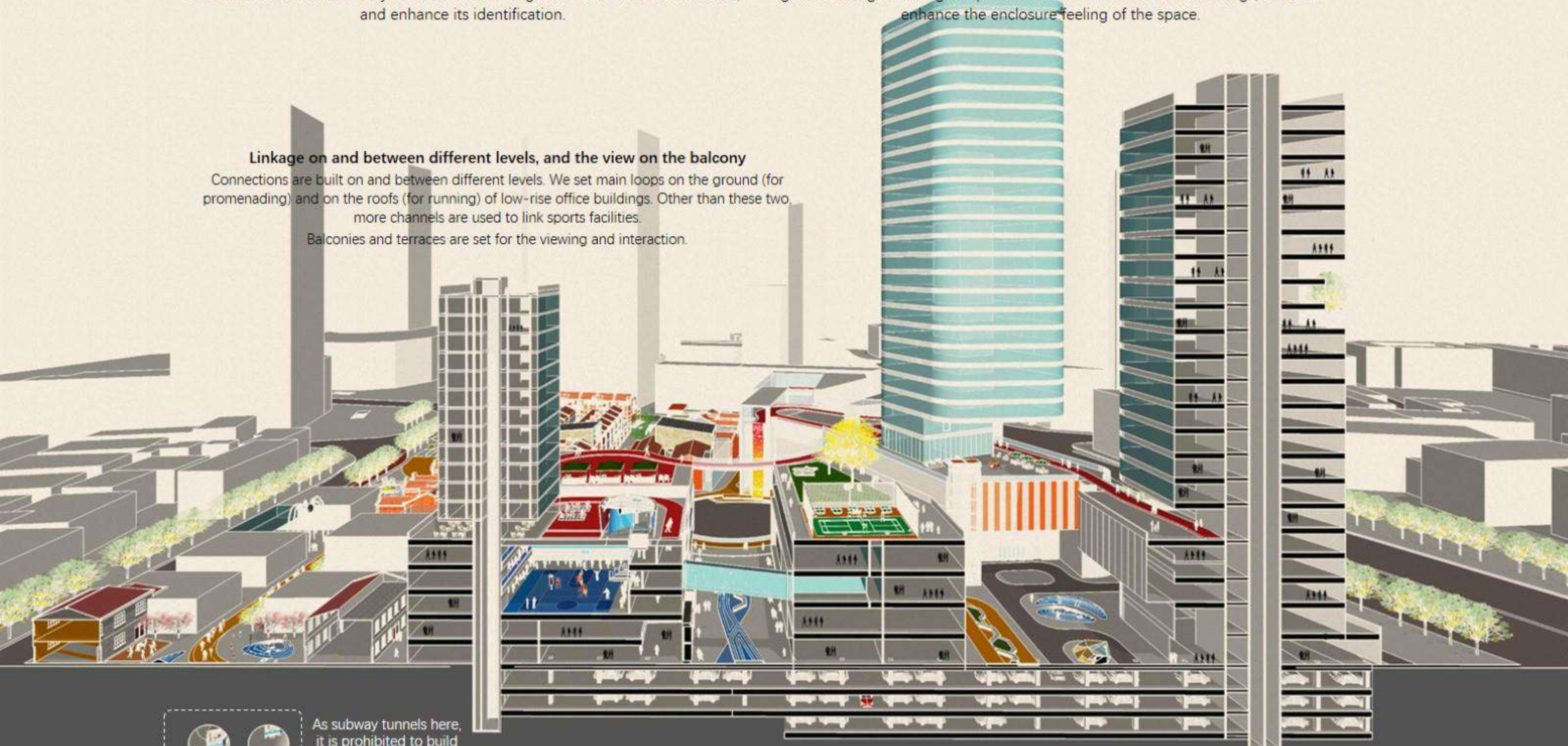
The bridge between buildings can be seen as a modern transformation of the old gate in Shanghai "lining". It promotes connection between buildings, and also enhance the enclosure feeling of the space.



#### Linkage on and between different levels, and the view on the balcony

Connections are built on and between different levels. We set main loops on the ground (for promenading) and on the roofs (for running) of low-rise office buildings. Other than these two, more channels are used to link sports facilities.

Balconies and terraces are set for the viewing and interaction.



As subway tunnels here, it is prohibited to build heavy structures above.





## U02 **LINKAGE & PLACEMAKING:** the next generation of railway station (area) in China?

Urban Design of X City Railway Station Area

Year: 2018-2020

Location: (confidential currently)

Type: urban design

Category: Real Projects

Role: collaborative work, responsible for research and placemaking work, as well as scheme design.

Railway station areas are often isolated from the city context due to the traffic segregation, oversize buildings and square in China. This project uses practical strategies to link all the elements, to downsize the space, to create a human friendly atmosphere, hence puts forward a new typology for station area practice.



### Site context

The current site is merely farm lands and has few elements and constraints, so the main challenges here turned to be dealing with railway station typology. Great potential should be released regarding to the advantage of mobility, land value after the station being built.



The current site is merely farm lands with few factories within them.



The railway tracks were surrounded by green space, which actually divided the areas beside.

Population density

Commute time

Land value

One major underground street was planned beyond the core area in the upper planning.



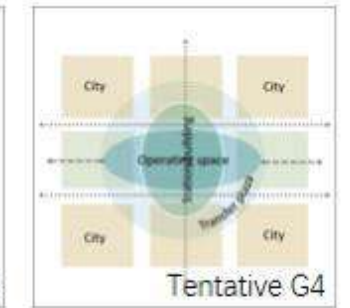
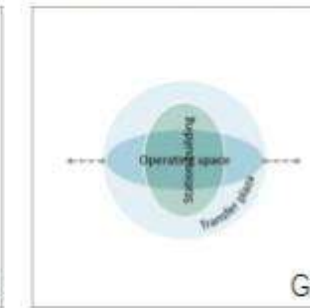
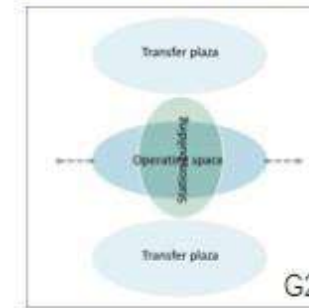
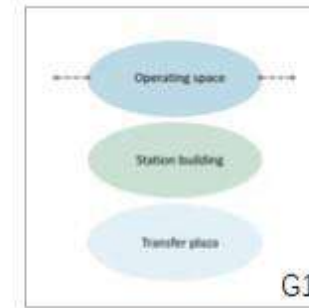
This site locates at a place with great mobility in the whole road network of Zhengzhou city.

- Core urban design area
- + Railway station
- Residence
- Greenery
- Office and commerce



### Stations Generations in China

Stations got compacted during the former generations, while we believe that, in the next generation, they should integrate with cities.



### Common issues for Chinese stations

Oversize squares with no vitality. Oversize station buildings block the flow. Mono-functional space with poor quality



### Chinese complexity: Social Context

Chinese railways are planned for long distance transfer, hence passengers often travel with plenty of luggage causing congestion, especially in holidays





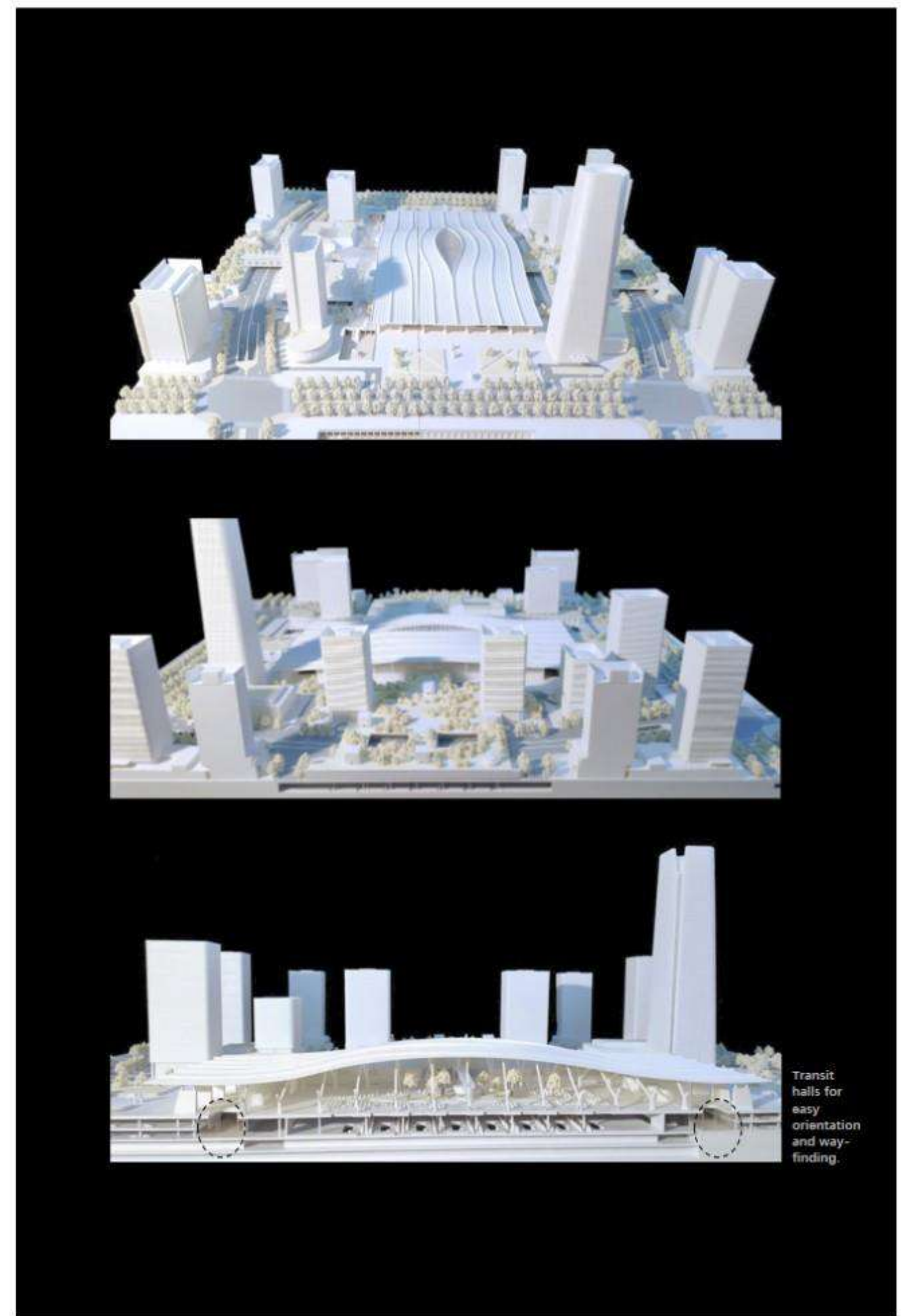


#### Strategies

1. bend the road to get more space for inner pedestrian-friendly development
2. Build garden above track and submerge fast traffics
3. Link all the districts surrounded
4. Use underground streets and plazas for to connect the city

#### Station areas for the normal and the holiday

In China, there is extreme high passenger flow during holiday like the spring festival, station area should be capable of this and should also be utilized in the ordinary time. The on-track gardens are places that can be used for the surround offices and residences, and for the holiday stranded passengers.







## Linkage & Level Changes

Linkages are created within and between different elements, with level changes. It brings not the convenience for movement, but also diversified experience in the space.



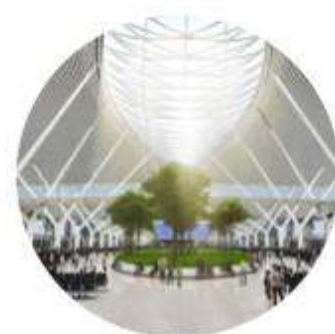
Bridges above the roads for pedestrians.



Walking commercial streets. Shopping when waiting for the train.



Provide more shaded places in case of the congested flow of passengers in holidays



The center hole on the roof lets in sunlight, and is visually identical



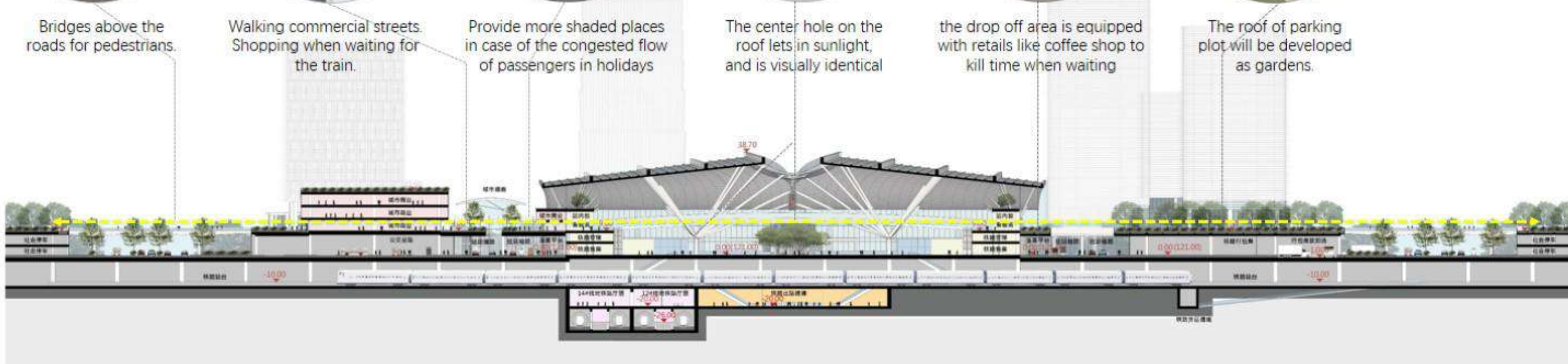
the drop off area is equipped with retails like coffee shop to kill time when waiting



The roof of parking plot will be developed as gardens.

## Linkage & placemaking

The linkages elements like bridges and platforms can be integrated into the space to create memorable landmarks and places.



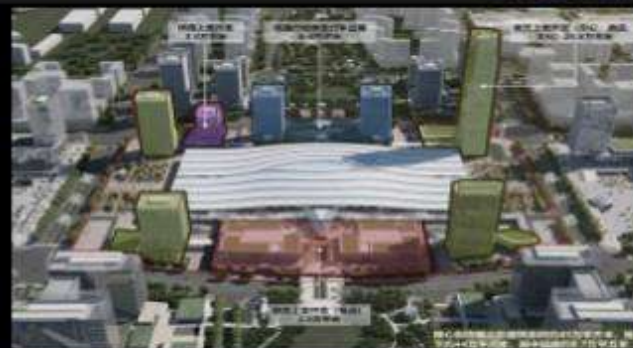
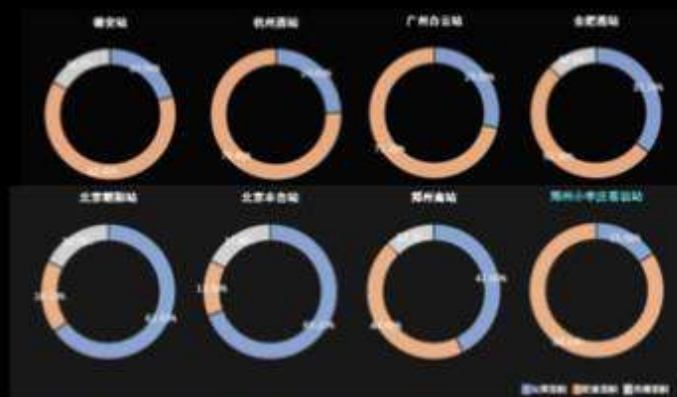
Section A-A'



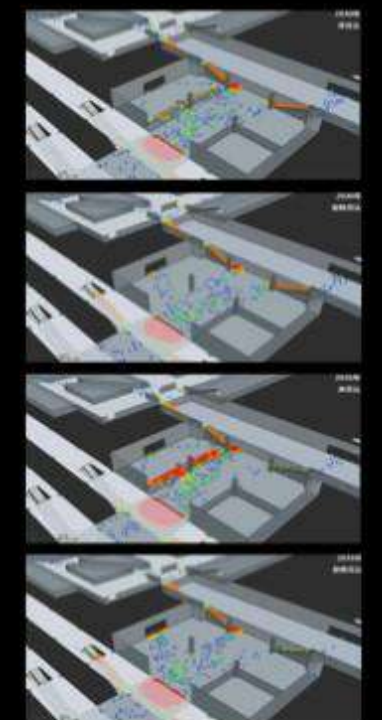
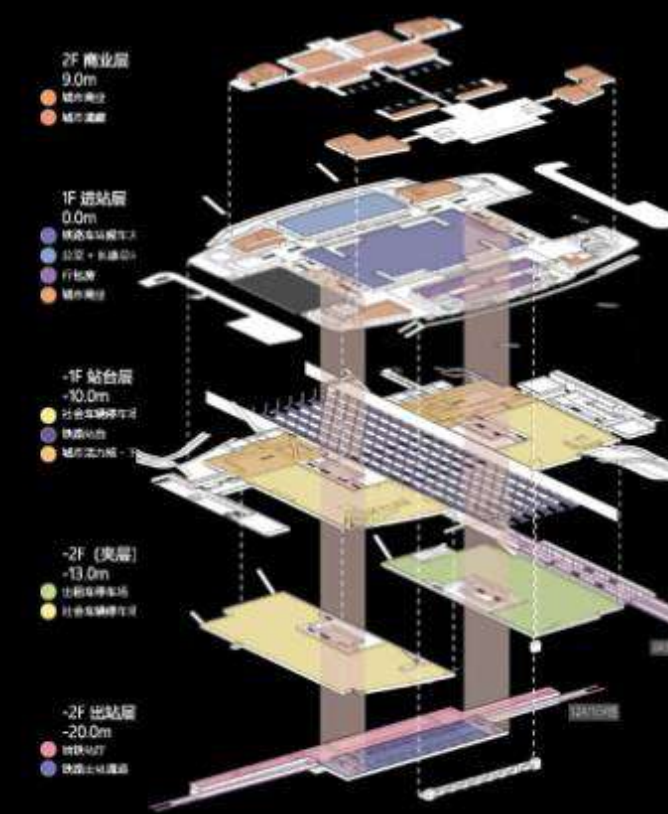
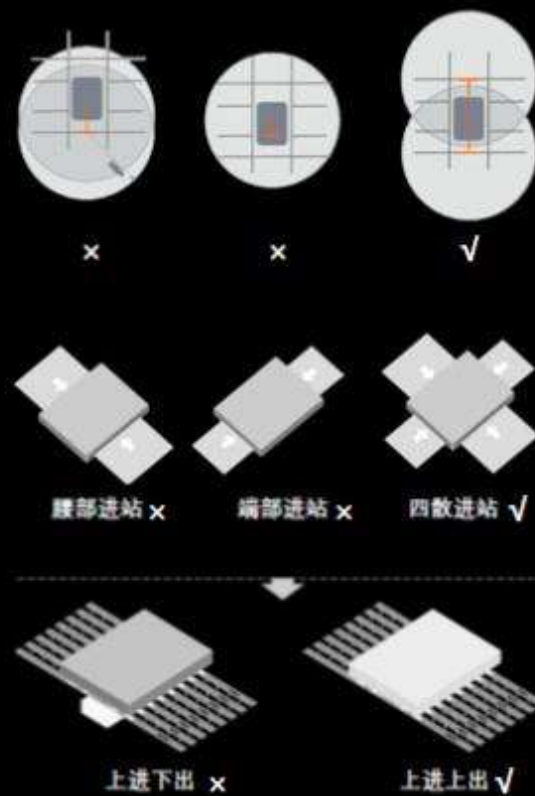


## Station building optimization

Optimization are conducted in terms of functions, circulations, considering the relationship between subway stations and the railway station, the entrance of the railway station, and the logistics and flows in the stations, etc.



we provide more city facilities and property than normal for better station-city integration



The flow simulation for the transit center hall area





## A01 SEE & BE SEEN

An office building design in Hefei

Year: 2017

Location: Hefei, Anhui, China

Type: architecture design

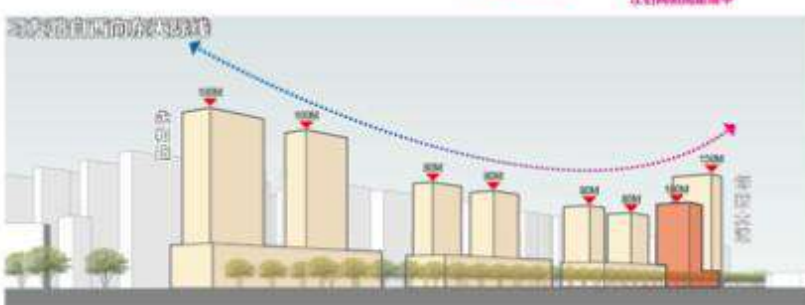
Category: Real Projects

Role: collaborative work as leader of 3 members

In this project, building is shaped by the surrounding buildings with which the street skyline is structured continuously. To maximize the view for the office building itself, several open space is insert for sharing. Efforts have also been done to provide the roof of the low-rise part building to the nearby residential district as a compensation for the occupying of its' original green land.







### Be seen

building is shaped by the surrounding buildings with which the street skyline is structured continuously.



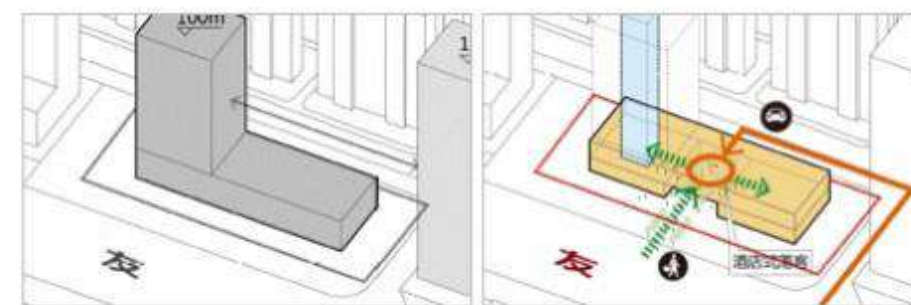
### See

To maximize the view for the office building itself, several open space is insert for sharing.



### Compensate the green land

Open the roof to the public as a compensation for occupying their previous green land.



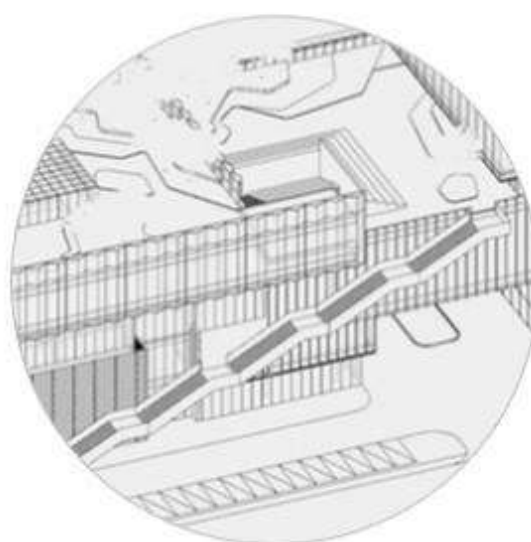
### Channel the street wall

Separate the low-rise part into two to make it more accessible for pedestrians, as well as to easy the ventilation of logistic.



### Provide "Secondary floor" in the sky for the office

Make it feels like living and working on the ground, where nature is nearby.



### Community-friendly

Facades are design with human-scale sizes and diversity. A grand stair is set to make the roof garden accessible to the public. Several memorable places are designed for interaction and fun.



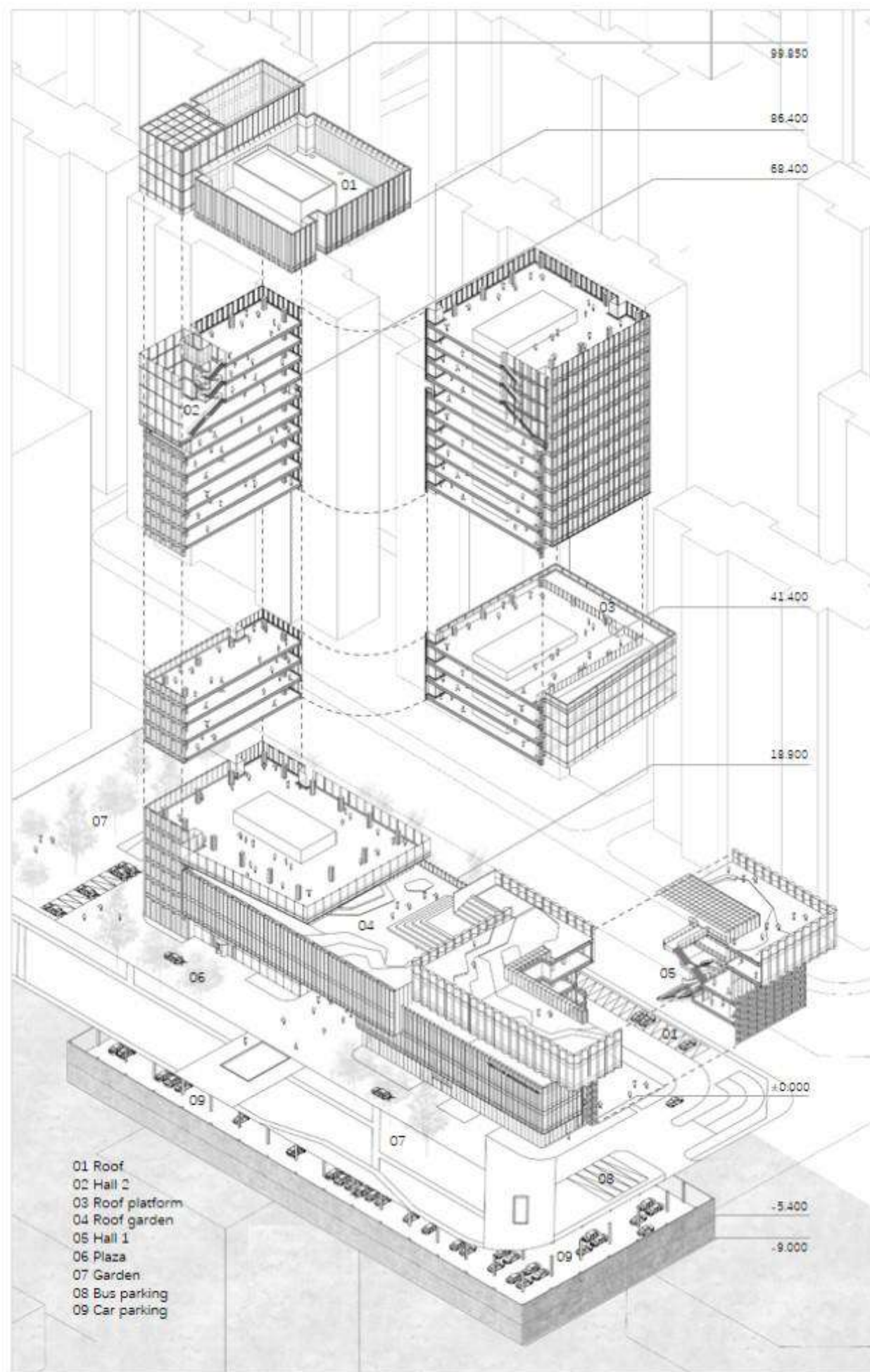


### Public spaces as energizers

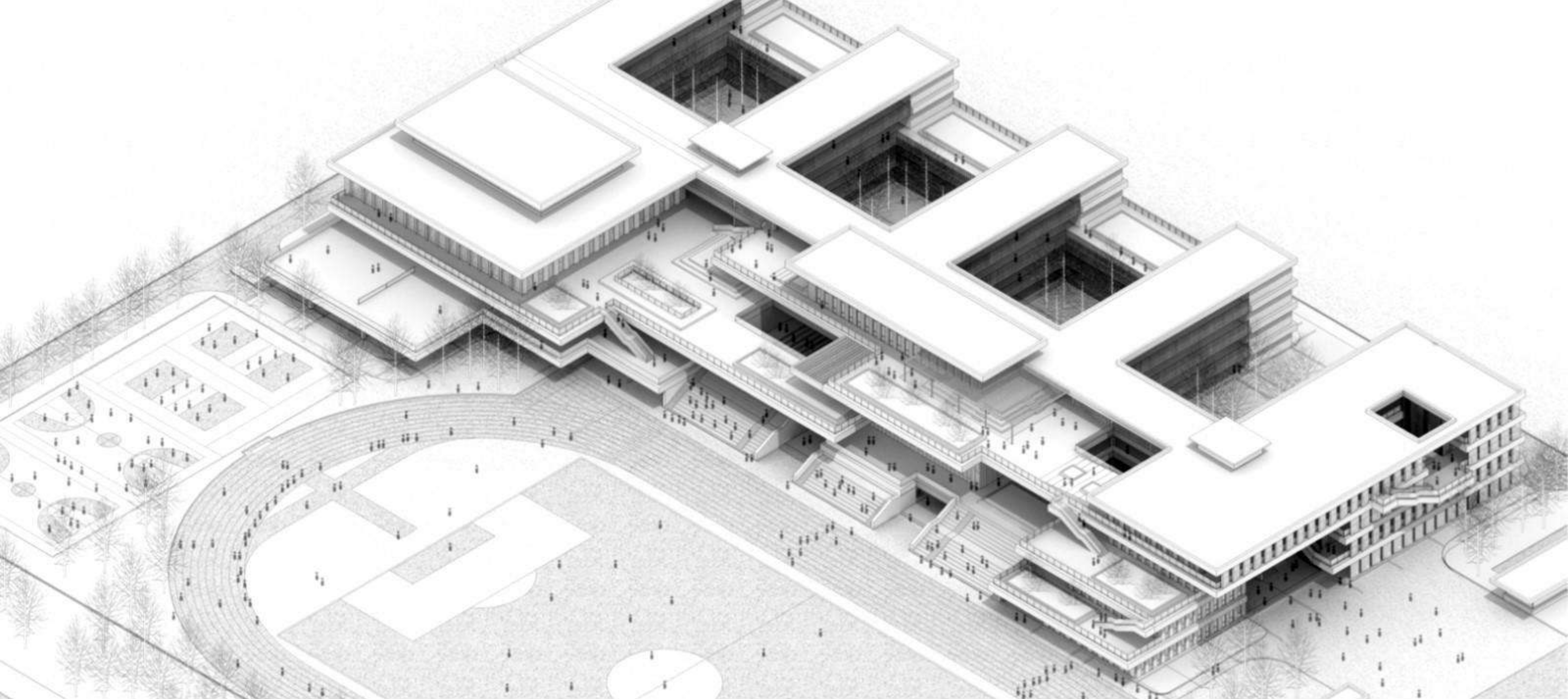
We set three main public space which is resulted from view analysis, to augment the charm these three spaces bring, activities and experiences are further designed and refined. New technologies for sustainability are used.











## A02 PLATFORMS & COURTYARDS

A secondary school design

The first prize in this project design competition

Year: 2018

Location: Hefei, Anhui, China

Type: architecture design

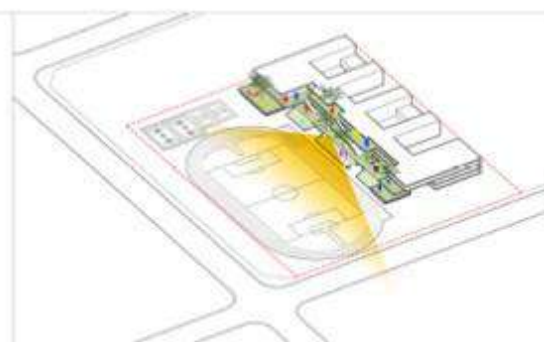
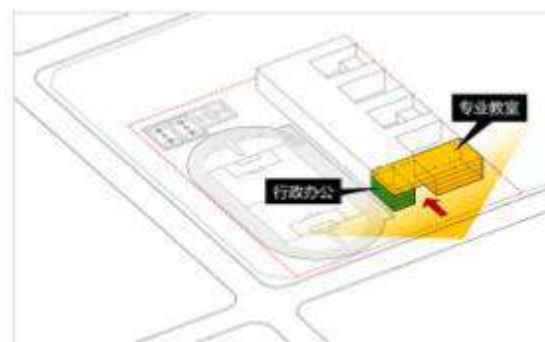
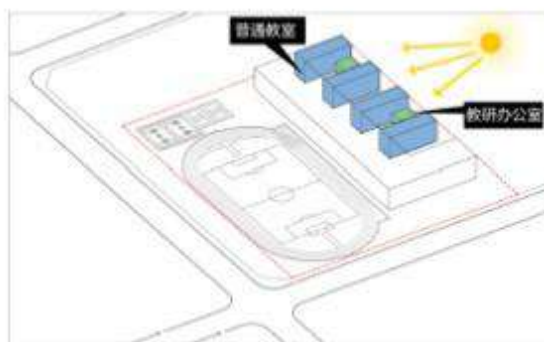
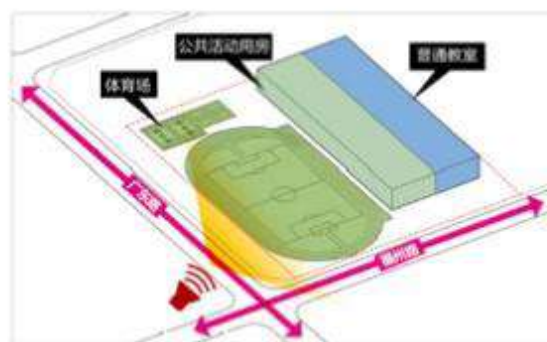
Category: Real Projects Competition

Role: collaborative work, as the leader of 3 members.

For a school, we want provide as much nature as possible to the students, especially when nowadays children are kept in the building all day. So we created the platforms. They can be used as grandstand, terraces for class breaks, and place for communication, etc. the building also got it landmark vision through this.





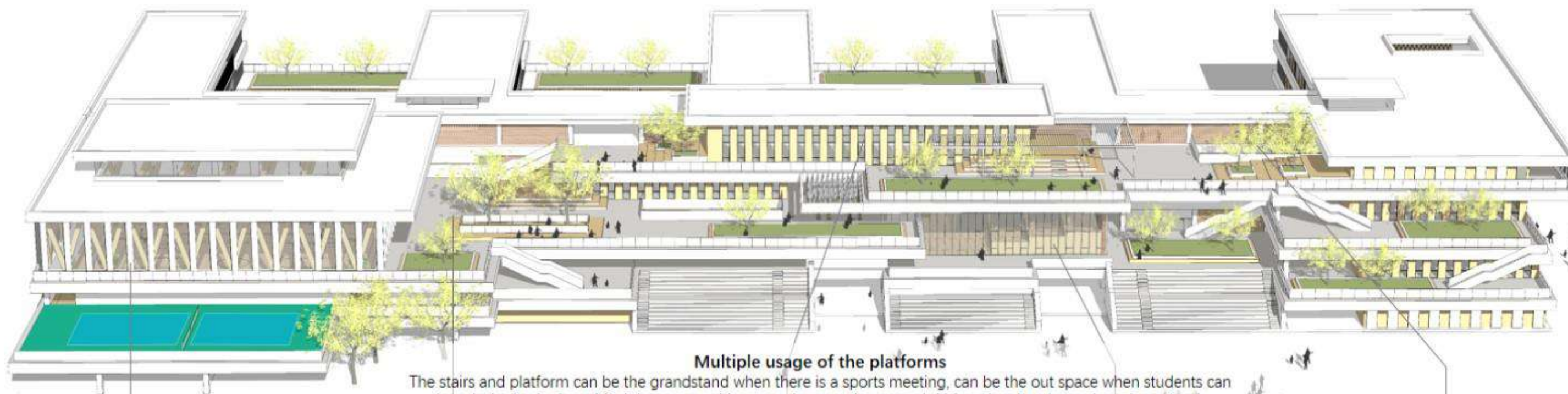


**Step1:** set the playground, avoiding noise from the traffic.

**Step2:** set courtyards, letting the sunlight in.

**Step3:** set the office building and practice classrooms as the gate of the school.

**Step4:** set the multiple activity room roofs as the public space.



### Multiple usage of the platforms

The stairs and platform can be the grandstand when there is a sports meeting, can be the out space when students can breath the fresh air and feel the nature without go down to the ground during class breaks, and can be a "book mountain" where they exercise physically. This is a place full of diversity and possibility.



Stadiums are constructed with some well-design big structure, we use these as the interior decoration.



Platform planted with trees can be a place like the ground. Children feel the nature here.



The long corridor is an open public space for students in different grades to communicate.



The library isn't only about books. It has stair, big roof, and small boxes where children can read and play.



Courtyards within platforms let the sunlight in, and the shadows play. Make platforms places of viewing.

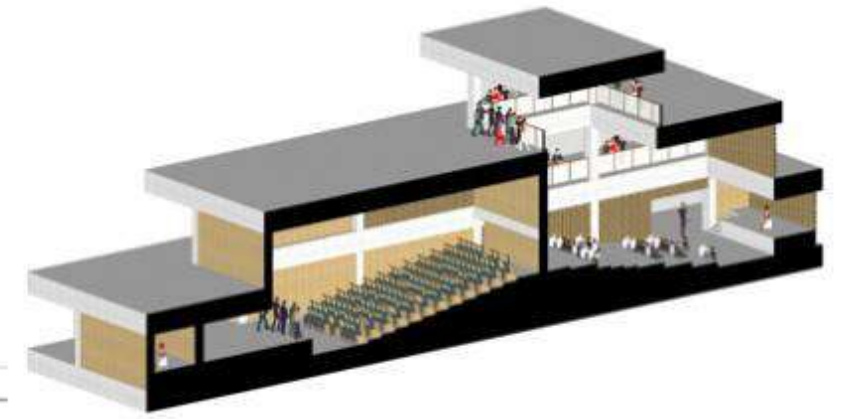




The building stands like a gate. The space under the big roof is both interesting and functional. It can be the place for parents waiting their children, or the stairs that children sit on and play with.

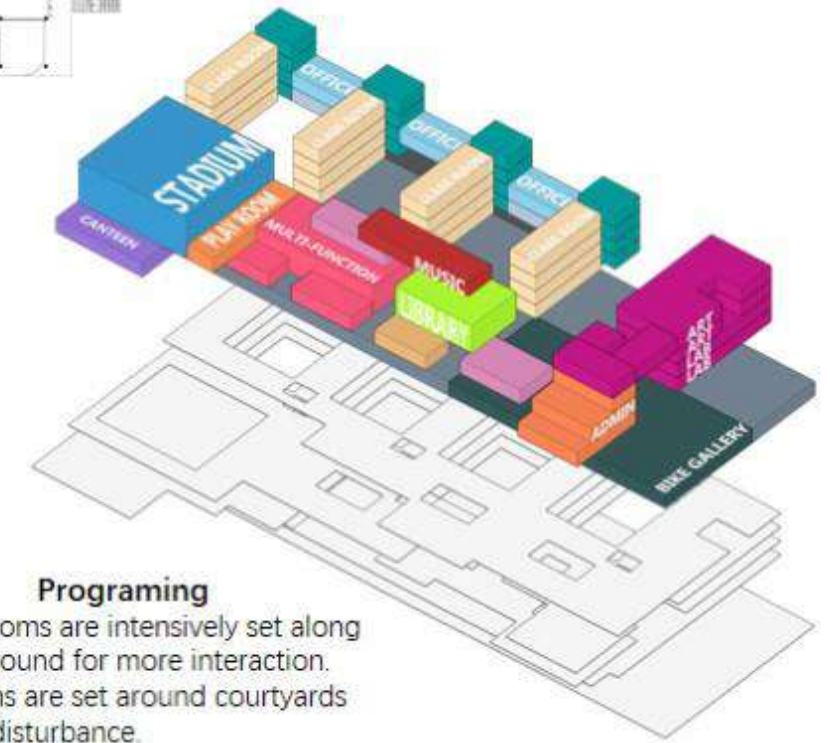
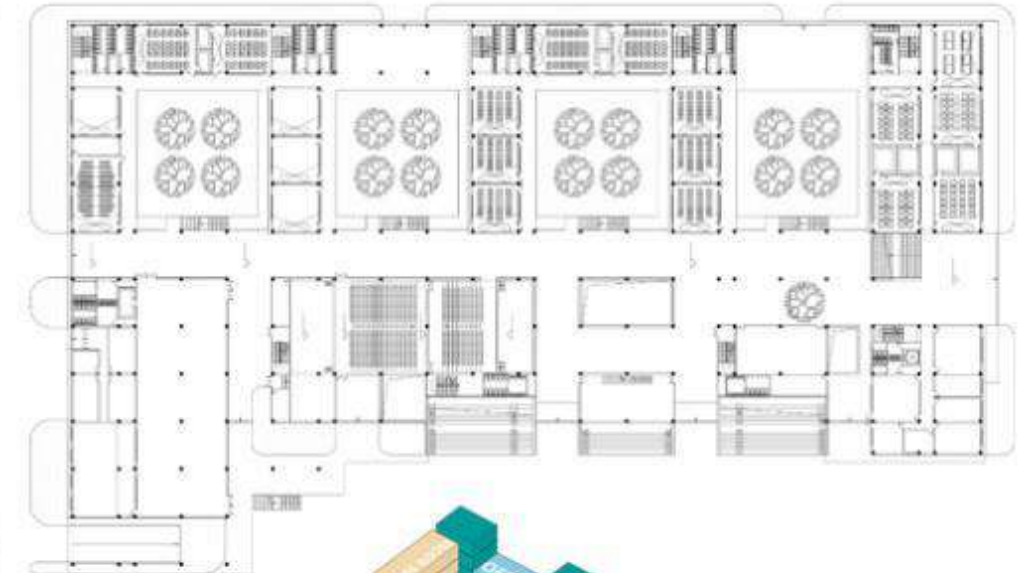
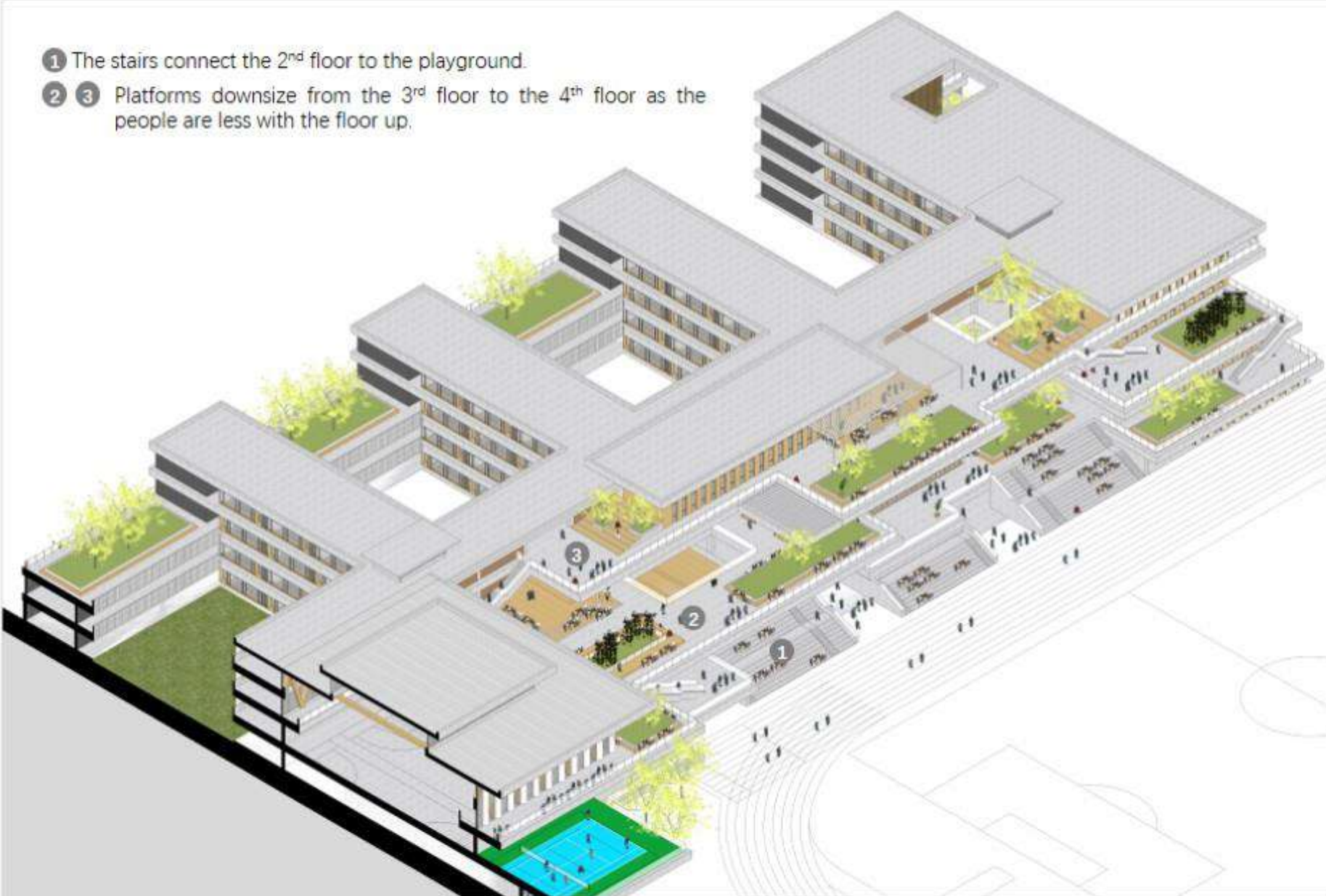


A traditional painting series "梅兰竹菊" is used as the theme for the courtyard. It enhance the atmosphere of art, education, and culture.



We set a stair behind the auditorium as a public space and lift the roof. The form change in height is a hint of space hierarchy. And the level change brings activities of seeing and be seeing.

- ① The stairs connect the 2<sup>nd</sup> floor to the playground.
- ② ③ Platforms downsize from the 3<sup>rd</sup> floor to the 4<sup>th</sup> floor as the people are less with the floor up.



#### Programming

Activity rooms are intensively set along the playground for more interaction. Classrooms are set around courtyards avoiding disturbance.





## A03 BUBBLES & RINGS

Kindergarten

The second prize in the real project competition

Year: 2018

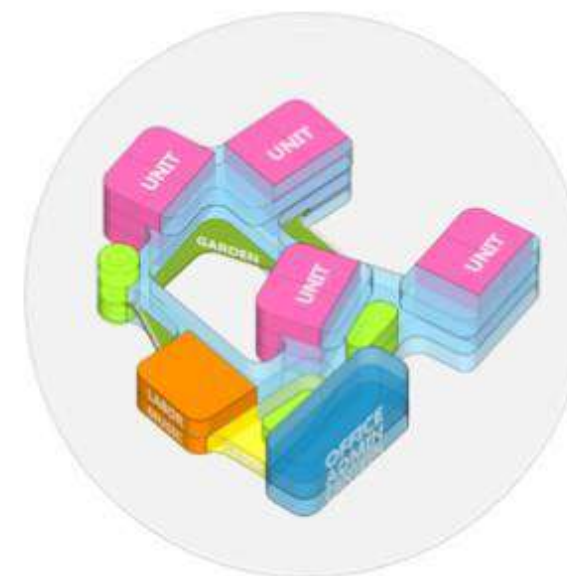
Location: Hefei, Anhui, China

Type: architecture design

Category: Real Projects Competition

Role: collaborative work, as the leader of 3 members.

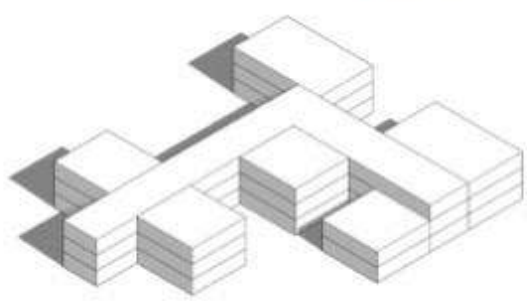
Inspired by the children's mischief, we tried to provide imaginative space and form in this projects through some twisting and transformation. Without fundamentally changing the layout, interesting terraces are created here. Some nodes as a stimulus are also added and connected in the relatively narrow site.



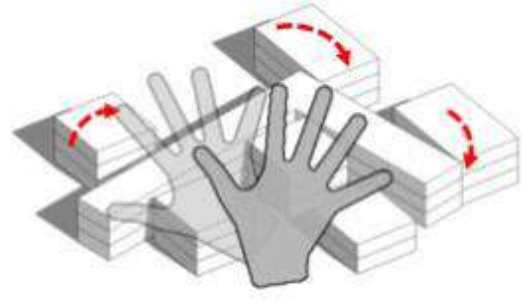


### Transformation for diversity

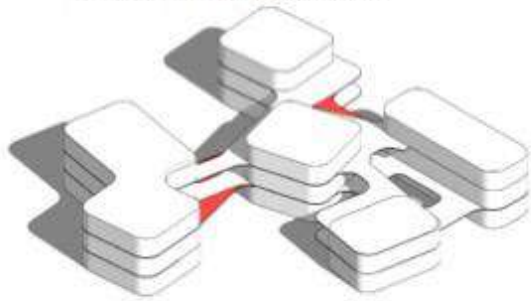
We mimic a child's naughty action of twisting the model, and get interesting terraces and corridor space.



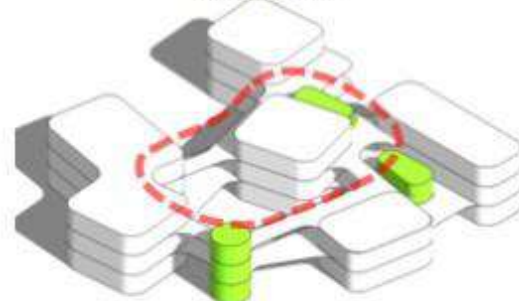
**step1.** A regular layout considering the sunlight, fire codes.



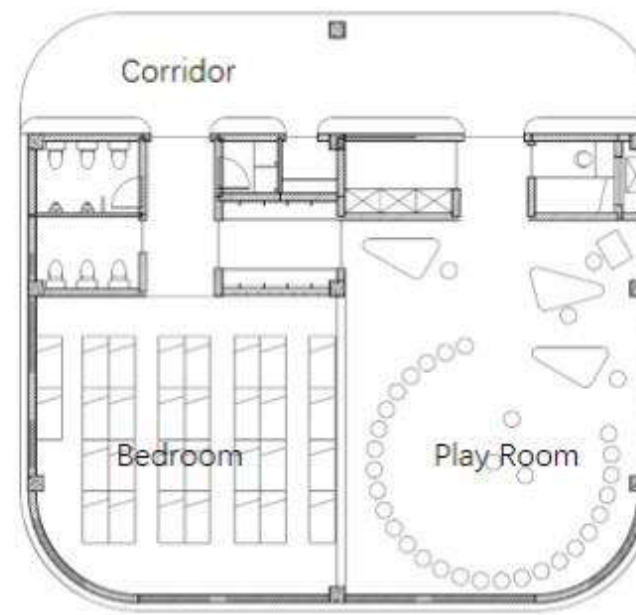
**step2.** When a kid twists the model



**step3.** After the twist, some interesting terraces occur

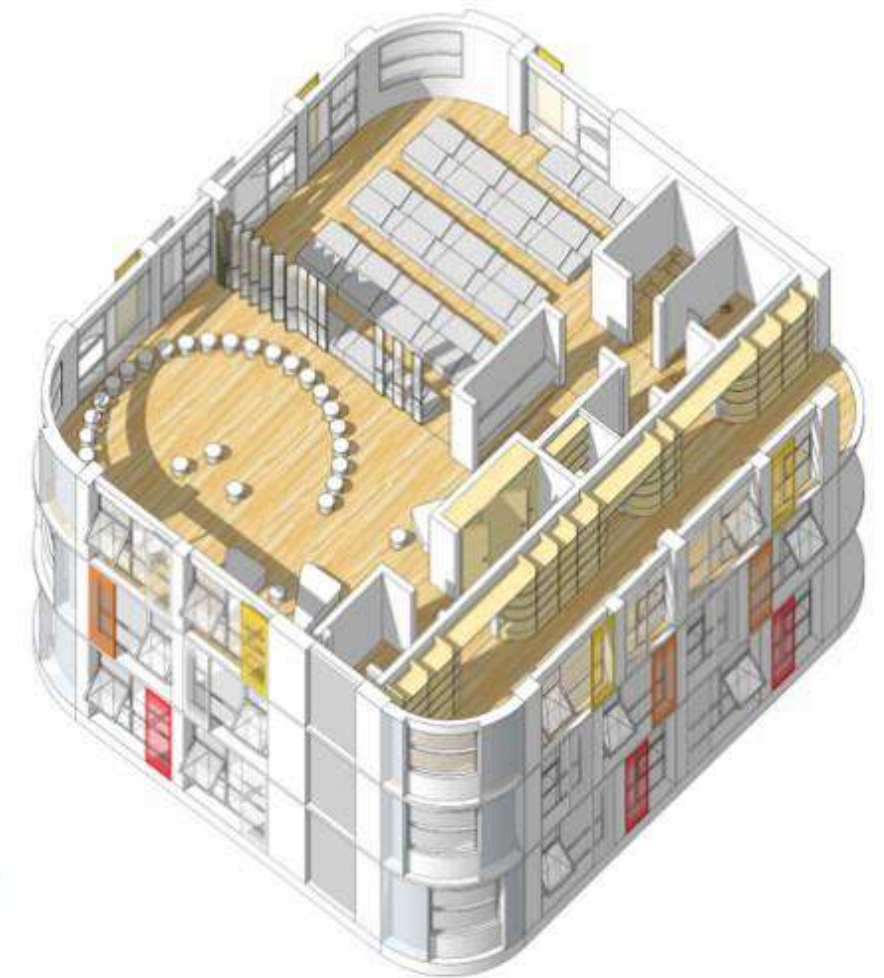


**step4.** Add some interesting nodes and connect them with a loop



### Unit design

We set curtains between the play room and the bedroom so that these two space can be combined when necessary. The corridor wall is been used as the case and showcase for children's work. With woods around, this place is more like home than school.



The twist of corridor not only brings diversity to the space, but also intrigue children to explore everywhere of the whole building, as every corner is different from each other, and you can't see through the building at the corridor.







## O01 BOXES & TERRACE

A shopping mall and retail district design

Year: 2018-2019

Location: Jimo, Shandong, China

Type: architecture design

Category: real projects

Role: collaborative work as one of 5 members, involved in all the phases

Shop malls are normally huge with boring appearance in China. So we divided the whole mall into several boxes, and provided terrace as an adaption to the landscape. Hence made it fit into surroundings with diversity and activity.



## O02 AESTHETICS

A residential districts design

First prize in the real project competition

Year: 2017

Location: Hainan, China

Type: architecture design

Category: real projects

Role: collaborative work as one of 3 members, responsible for all the facades and the landscape axis.

Designing residential districts can be extremely boring in China. There is not much left for architects to create when real estate developers see these house as products only designed for easy construction and sale profits.

But in this projects, I found my own interests in the aesthetics of the facade, trying to make it look as elegant as possible, which turned out to be a successful outcome as the aesthetics brought commercial value to the "products".





大厅透视图



吧台透视图



洽谈区透视图



## 003 COLORS

A template Preschool Shop design

Year: 2019

Location: Shanghai, China

Type: interior design

Category: real projects

Role: individual work

A preschool education TINKAMO wants to set a template shop for its franchisees. There are two practical issues here, one is the visual identification of the brand, another is the design scheme which is supposed to be highly adaptable for different building plans. So here I use 4 colors to make distinct from other shops. And the form constitute is been made mainly on the surface: the walls, the floors, and the ceiling, which are free from the plan layout.



Standard colors

Plan

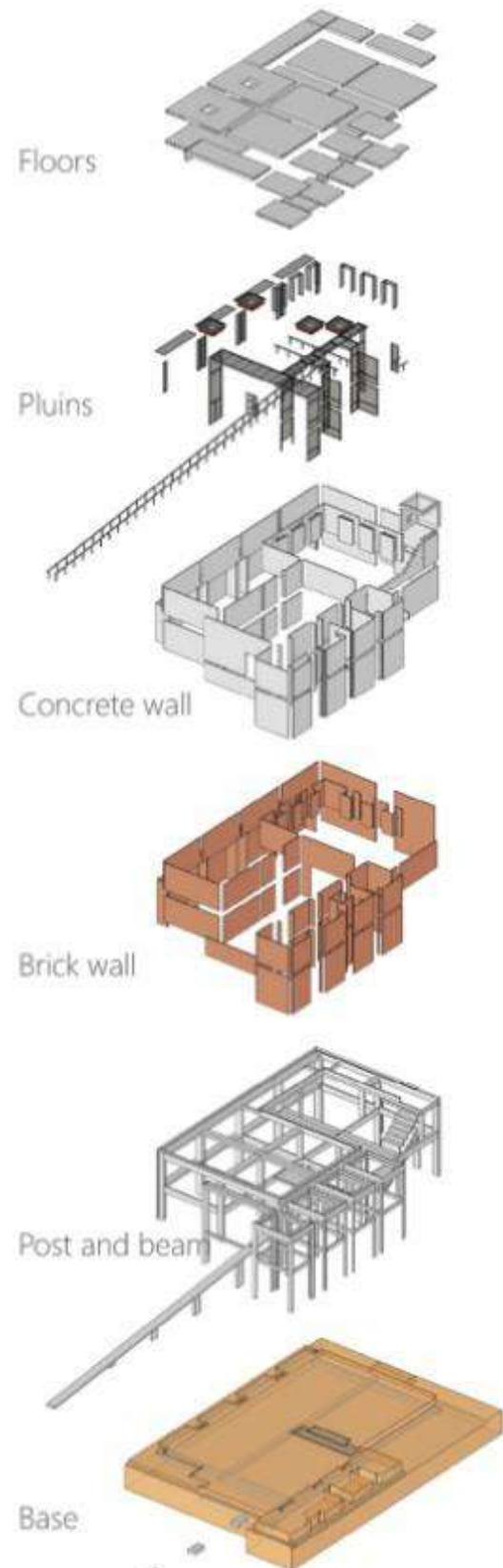
Celling





# O04 CRAFTMANSHIP An Analysis of Architecture Craftmanship

## structure



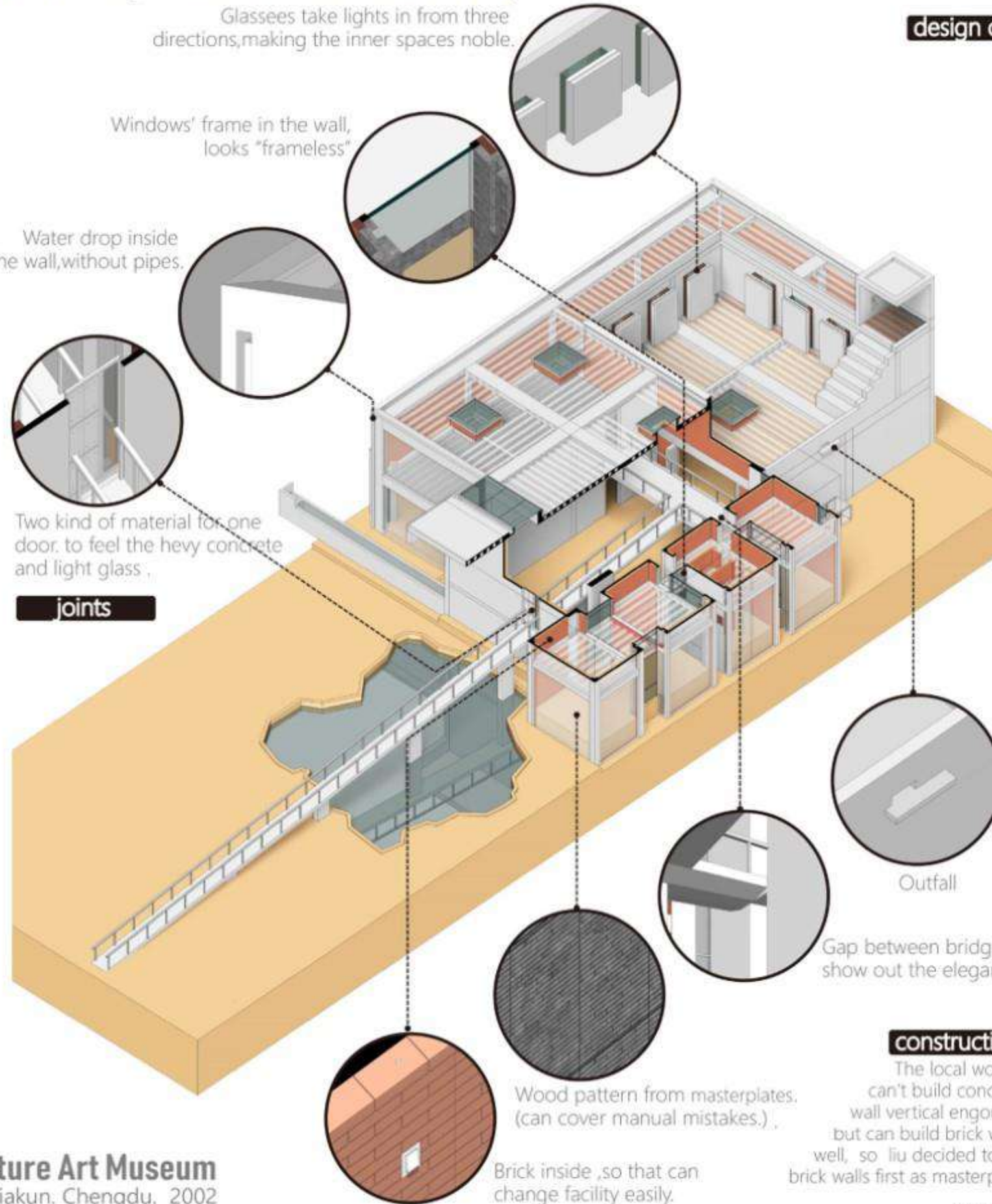
Glasses take lights in from three directions, making the inner spaces noble.

Windows' frame in the wall, looks "frameless"

Water drop inside the wall, without pipes.

Two kind of material for one door, to feel the heavy concrete and light glass.

## joints



## design continuity

Avioding being too casual, liu always use similiar crafts and architectonic details in his different projects, and update them.



Walls



Bridges



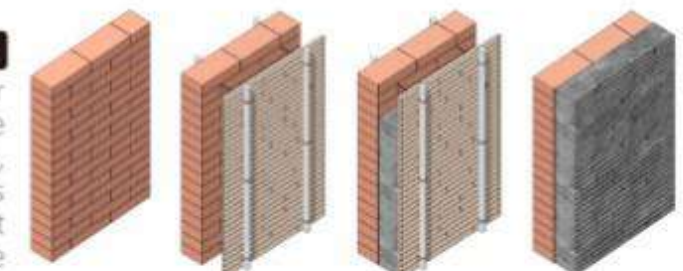
Patterns



Lights

## construction

The local worker can't build concrete wall vertical enough, but can build brick walls well, so liu decided to set brick walls first as masterplate



**LUYEYUAN** Stone Sculpture Art Museum  
Designed by Liujiakun, Chengdu, 2002

Name: Chenenshan (陈恩山) nationality: Chinese Student ID Number:1832199





# 书山叠景

金山区图书馆新馆 | 建筑概念方案设计 1



代尔夫特理工大学团队

余紫悦 徐子涵 田安安  
陈恩山 张宇卿 周正



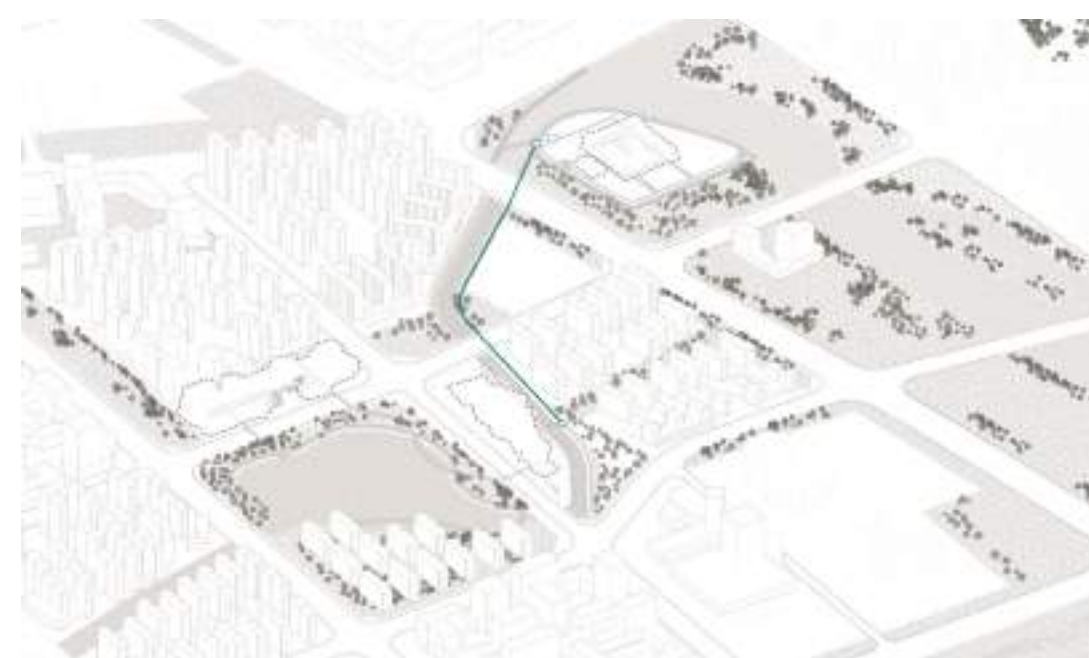
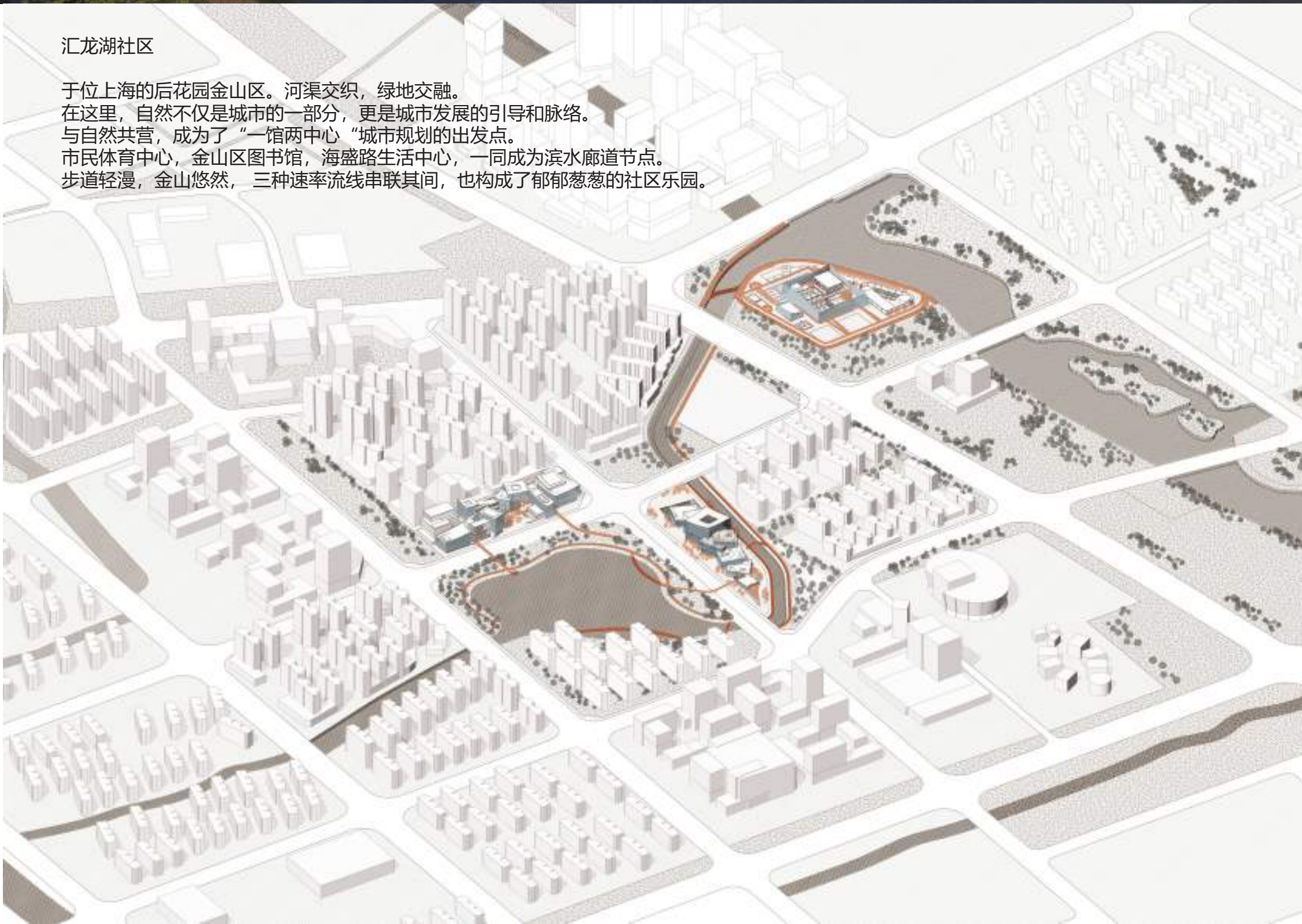
# 书山叠景

金山区图书馆新馆|建筑概念方案设计 2

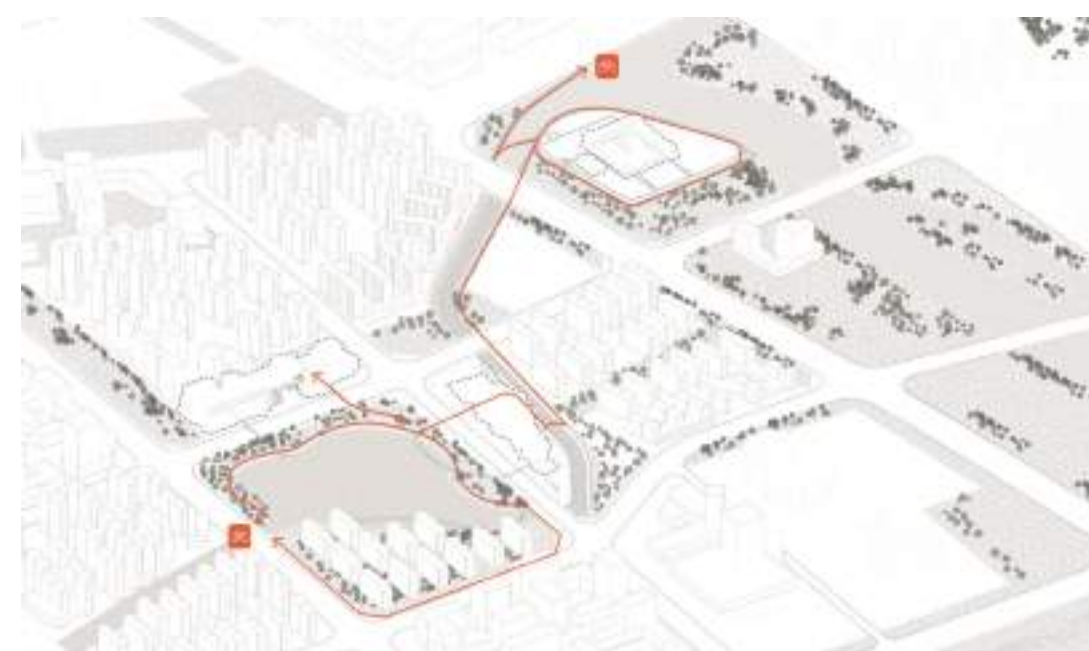


汇龙湖社区

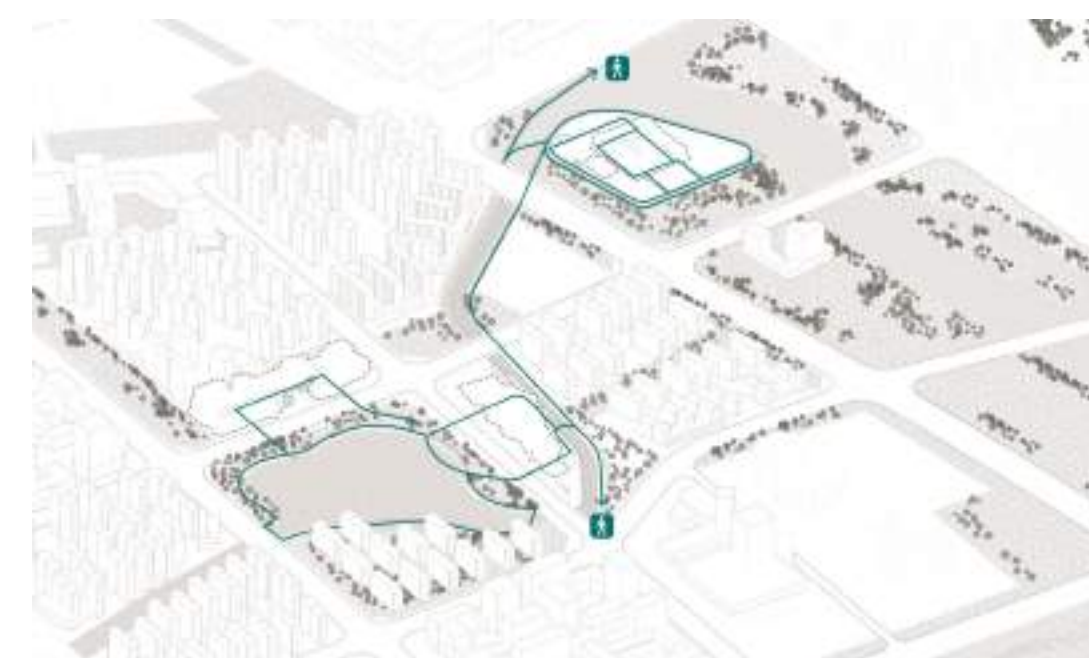
于位上海的后花园金山区。河渠交织，绿地交融。  
在这里，自然不仅是城市的一部分，更是城市发展的引导和脉络。  
与自然共营，成为了“一馆两中心”城市规划的出发点。  
市民体育中心，金山区图书馆，海盛路生活中心，一同成为滨水廊道节点。  
步道轻漫，金山悠然，三种速率流线串联其间，也构成了郁郁葱葱的社区乐园。



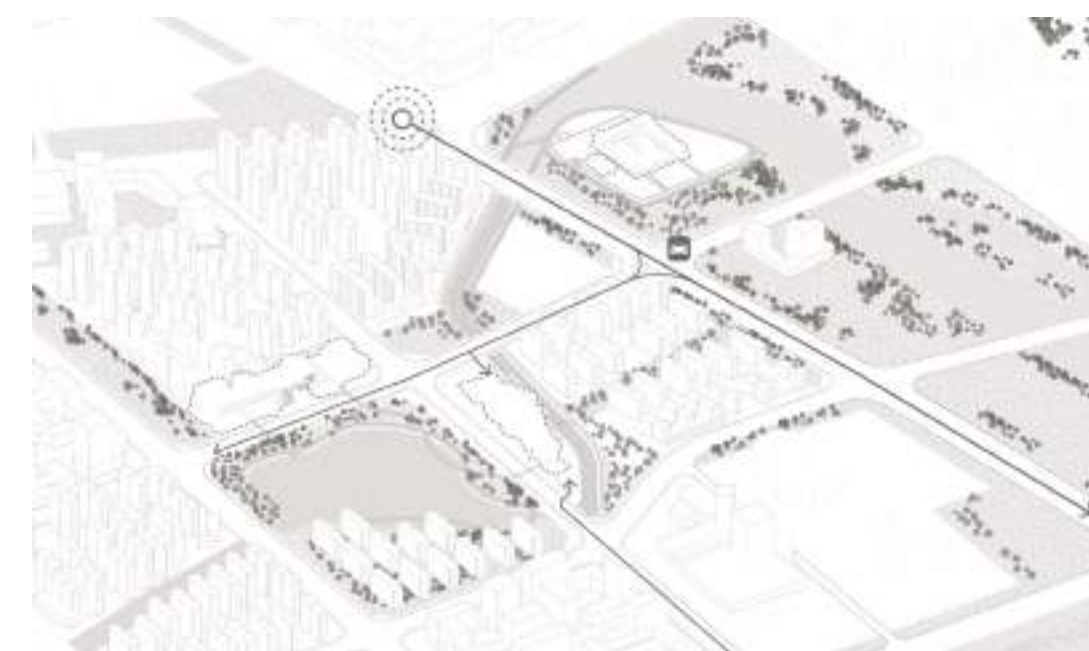
汇龙湖社区：500m生活圈



骑行系统：沿水岸享受骑行乐趣



步行系统：串接三中心与水网



道路系统：紧密连接周边路网



海盛路生活中心



市民体育中心



# 书山叠景

金山区图书馆新馆 | 建筑概念方案设计 3



## 方案生成



1. 碎化平面  
形成初始体量，在城市形成有机界面



2. 立体平台  
登高远瞰湖景，化解视线遮挡



3. 底层架空  
形成城市通廊，打通沿河城市绿带



4. 平台扭转  
丰富观湖视野，强化书山意向

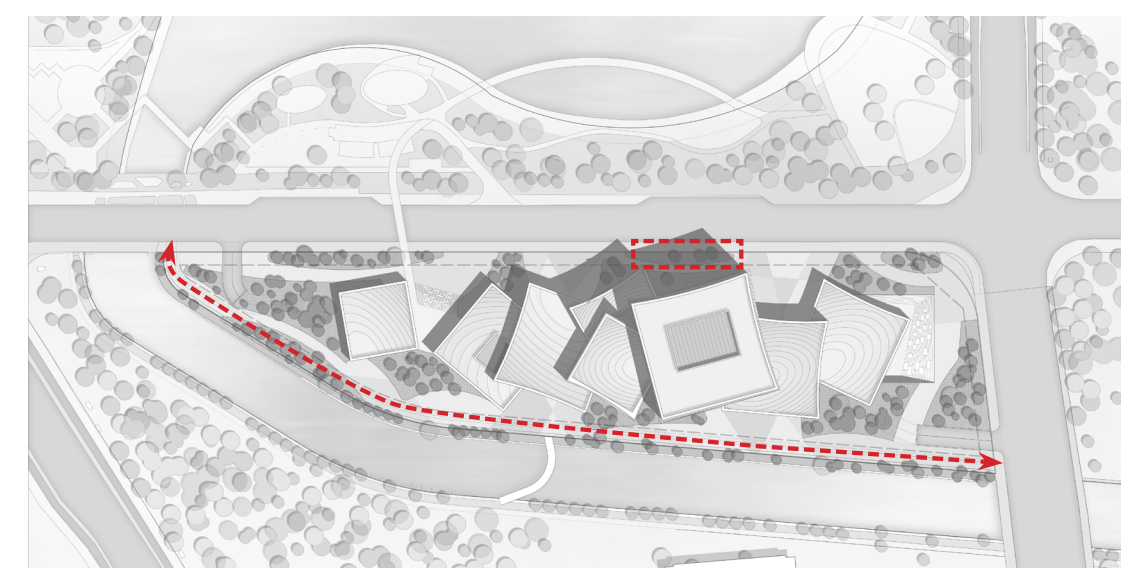


5. 形成体量  
置入功能，合理分区，形成叠落体量

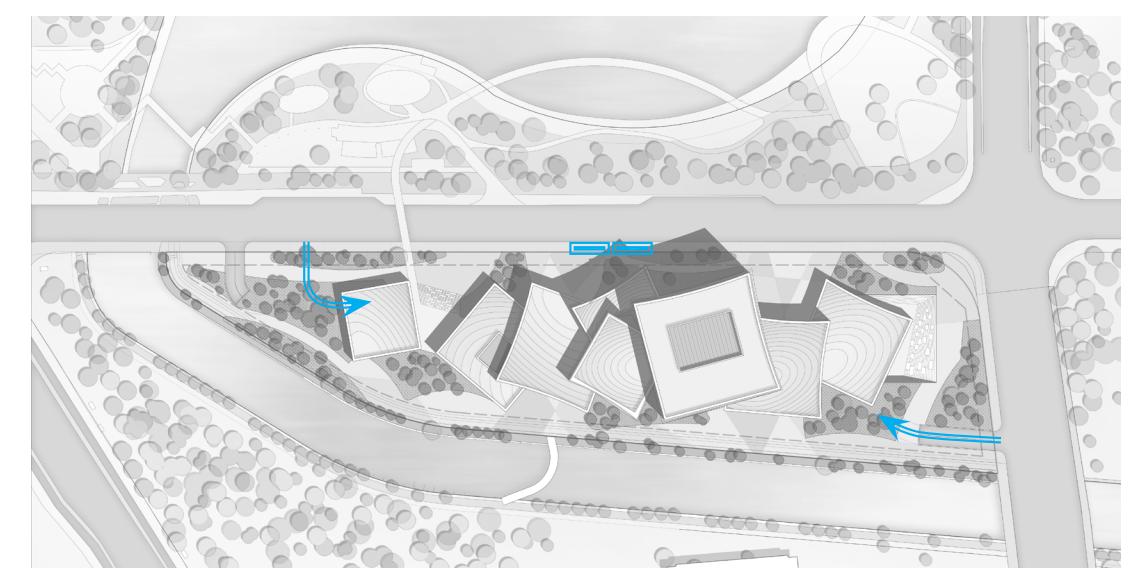


6. 活化表皮  
薄纱表皮微微掀起，显山露水

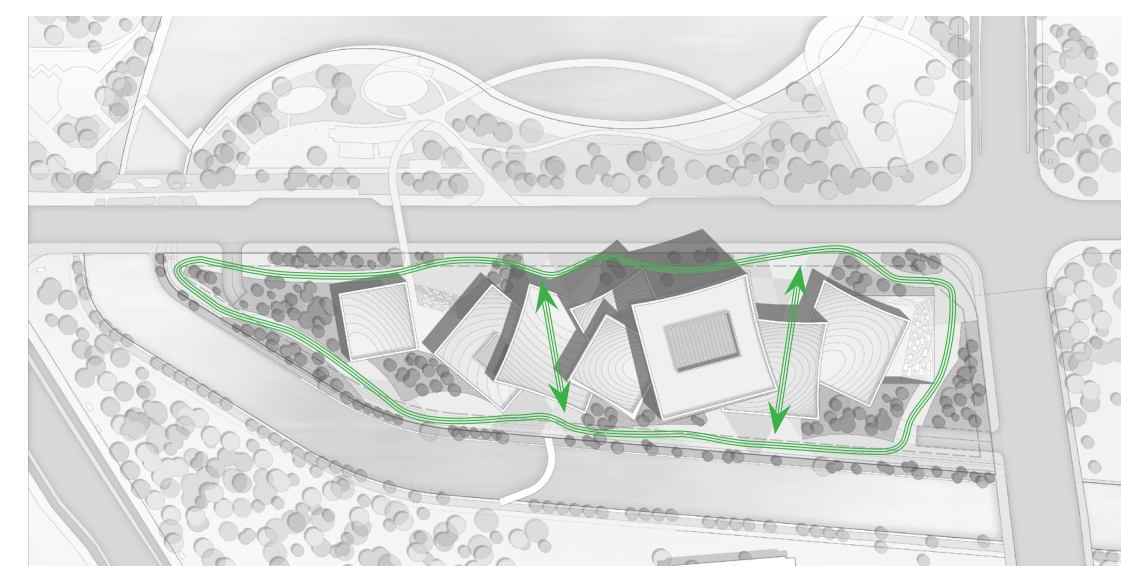
## 场地策略



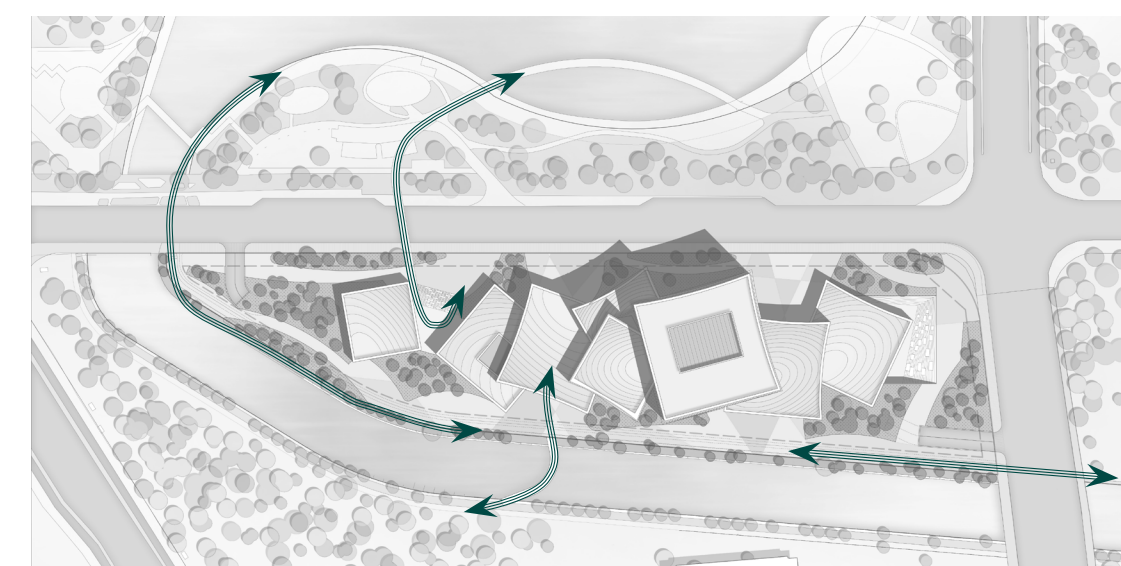
消防环道 & 消防登高场地



地下车库入口 & 大巴临时停车位

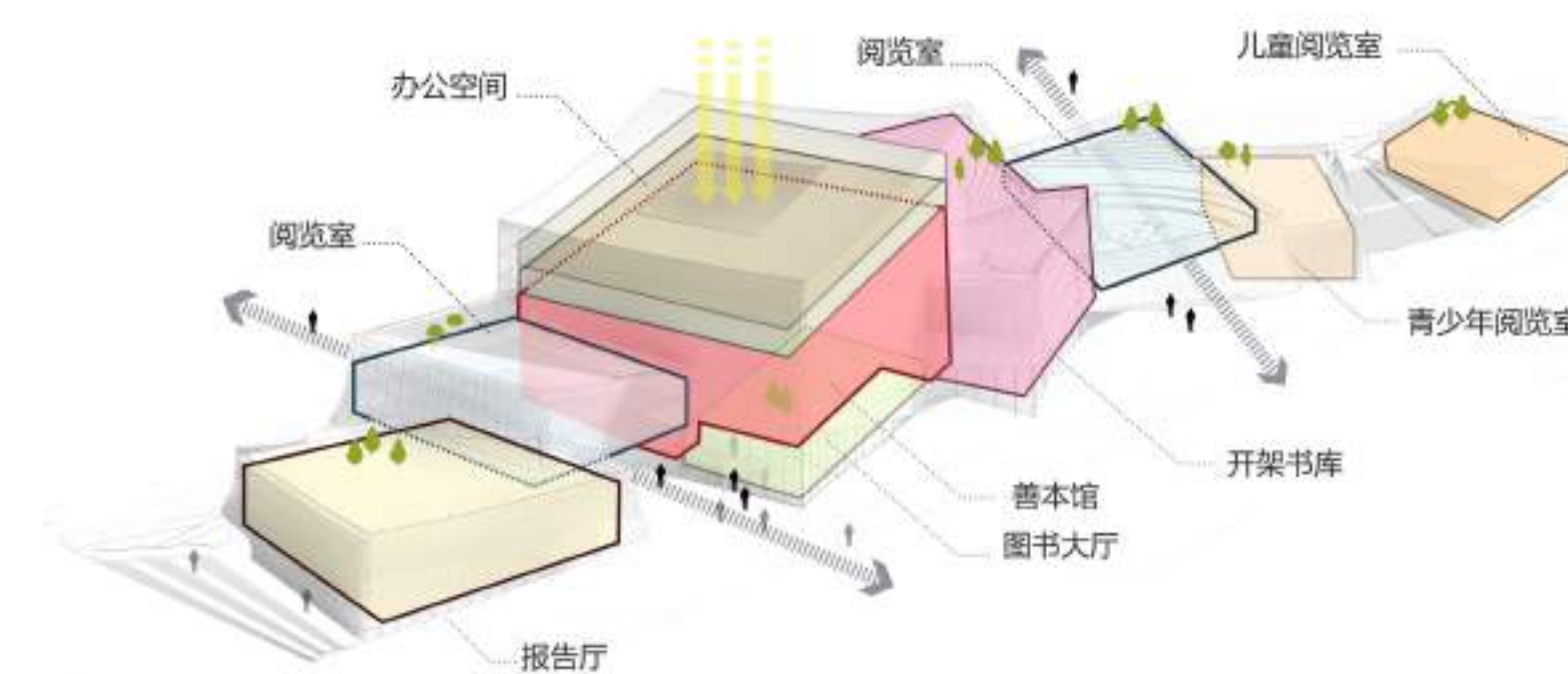


步行系统：场地内部



步行系统：与周边地块

## 功能量化

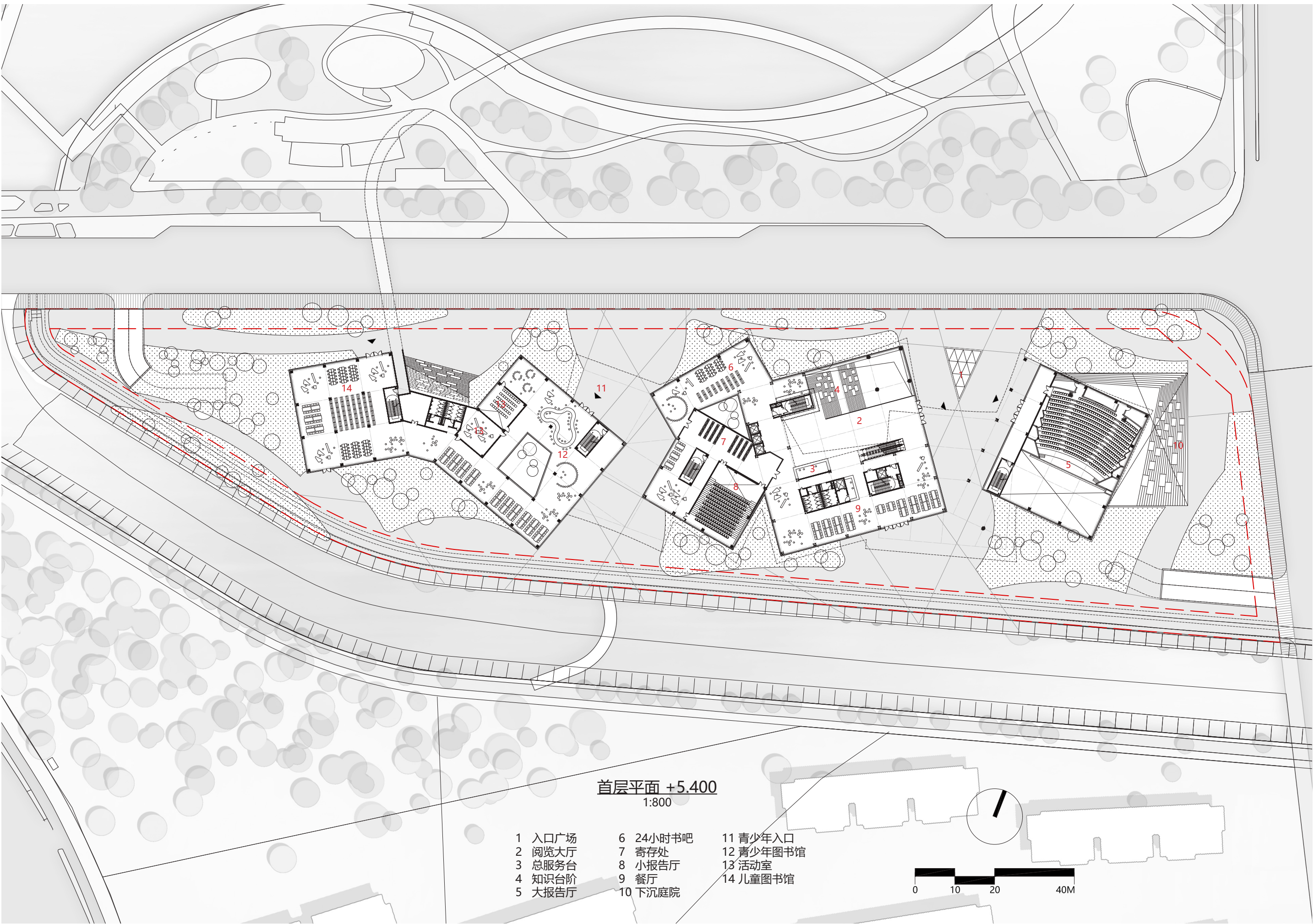
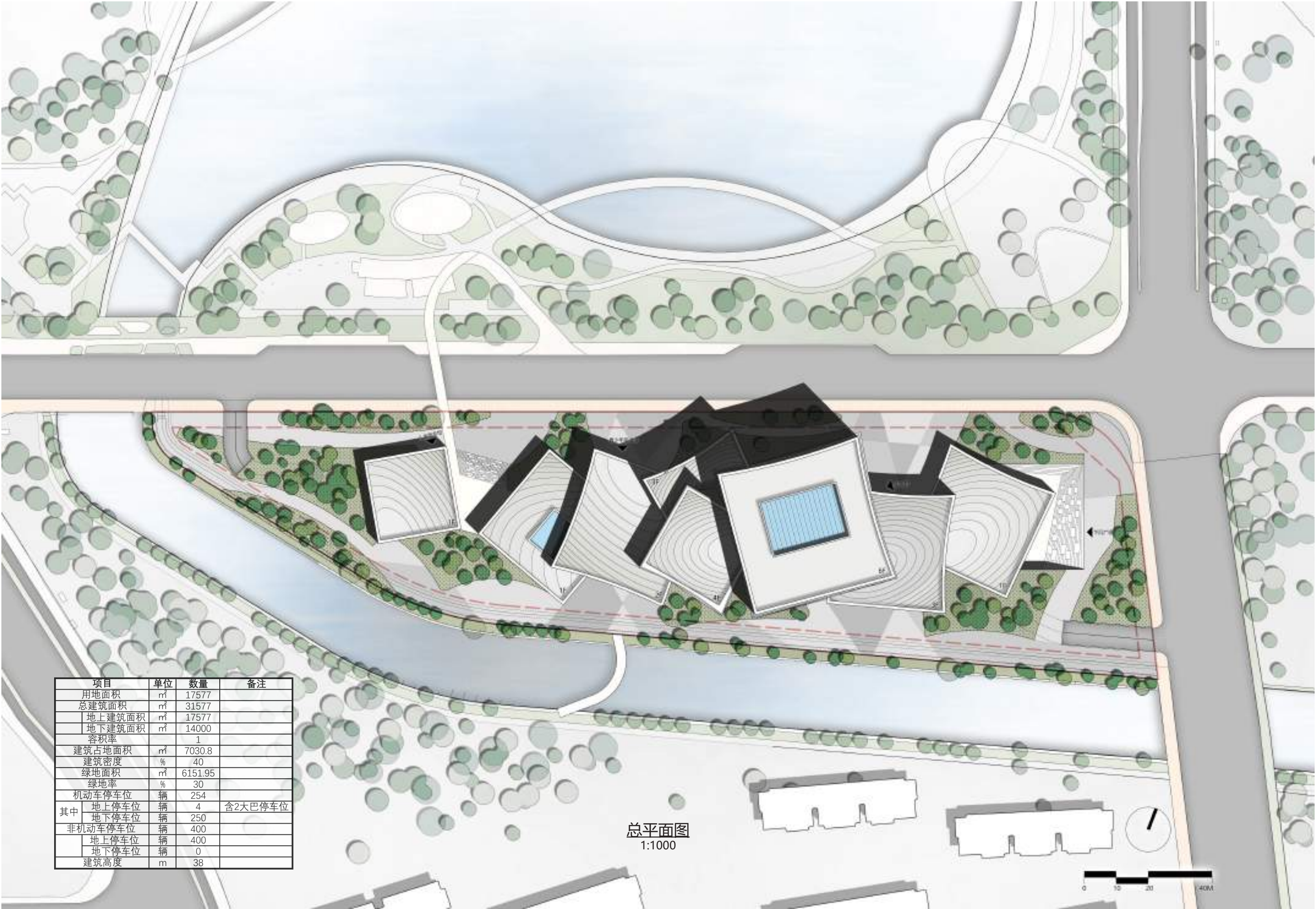




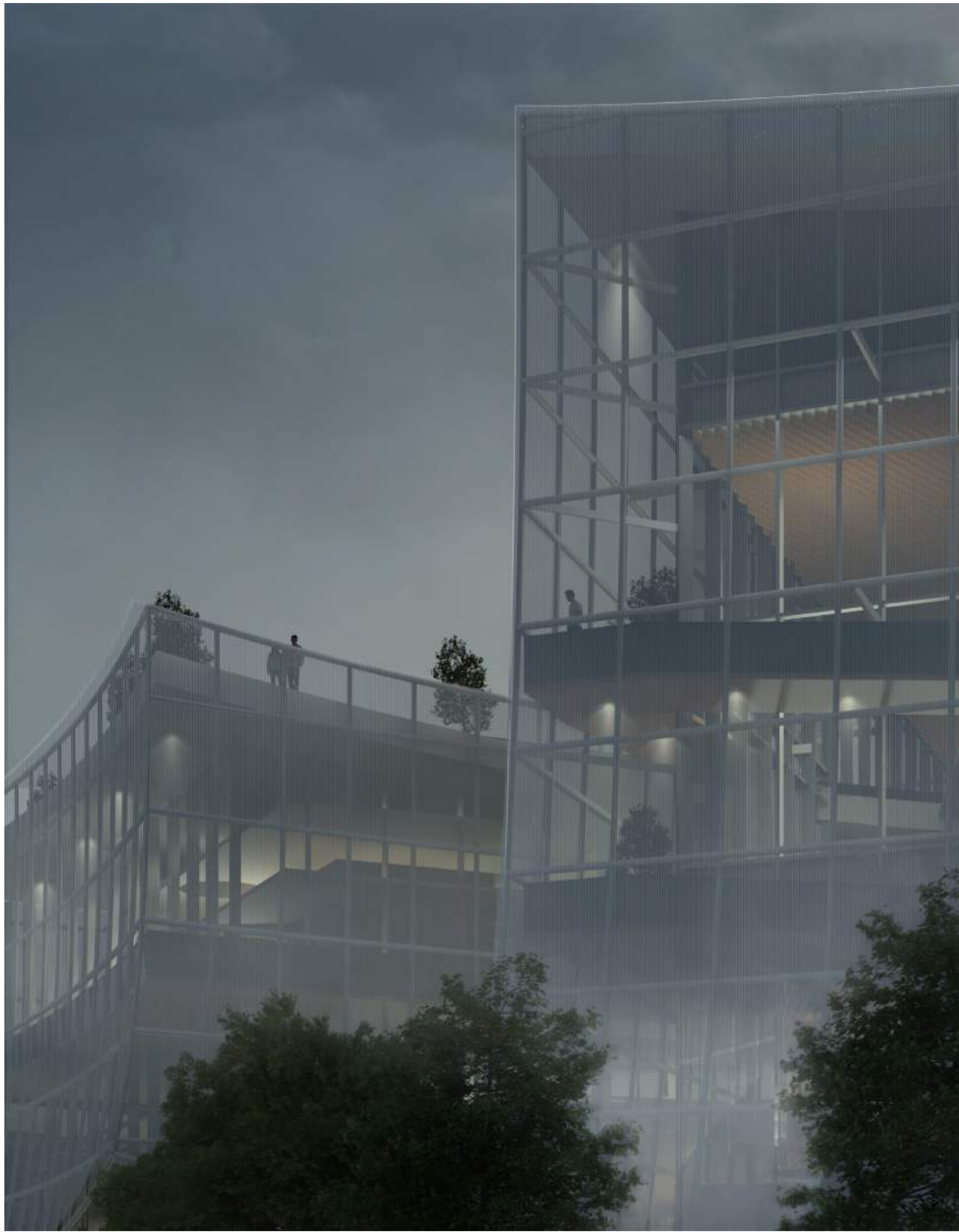


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# 书山叠景

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## 通高中庭

与双层玻璃幕墙引入丰富的自然光。  
为室内打造开阔，明亮的多功能城市会客厅。  
屋顶花园浸入阅读空间，模糊了室内外的边界。

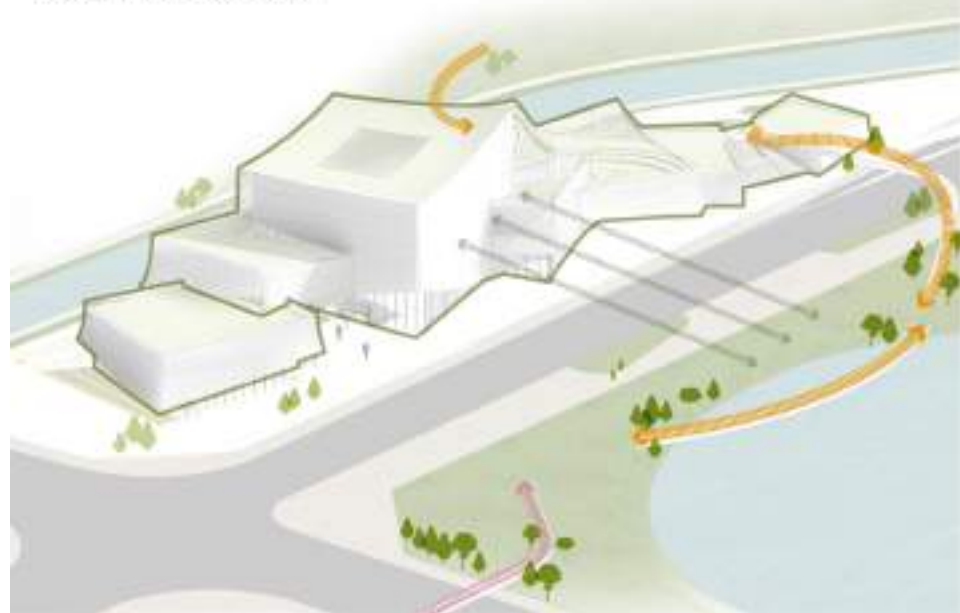


## 空间策略

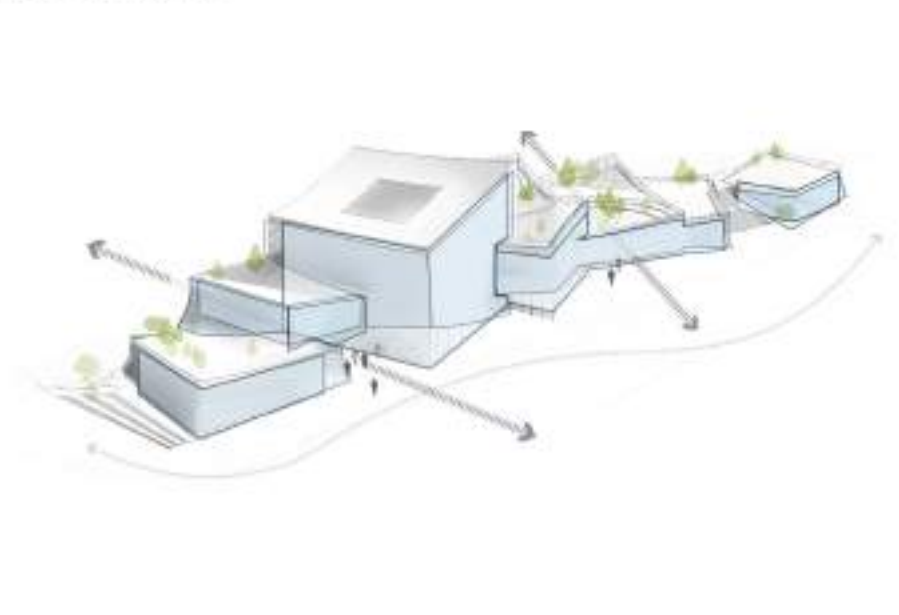
周边景观视野最大化的建筑形态



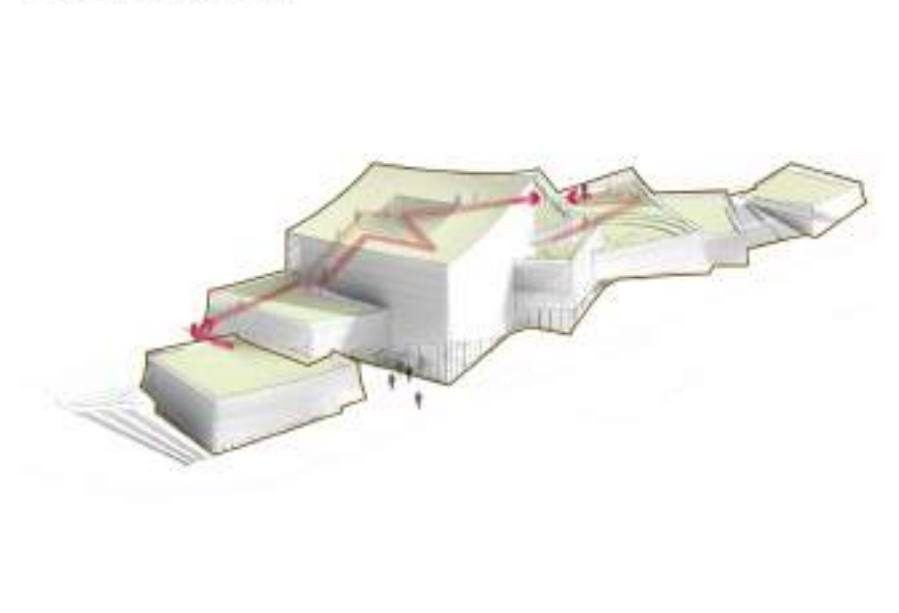
与场地景观融为一体生态廊桥



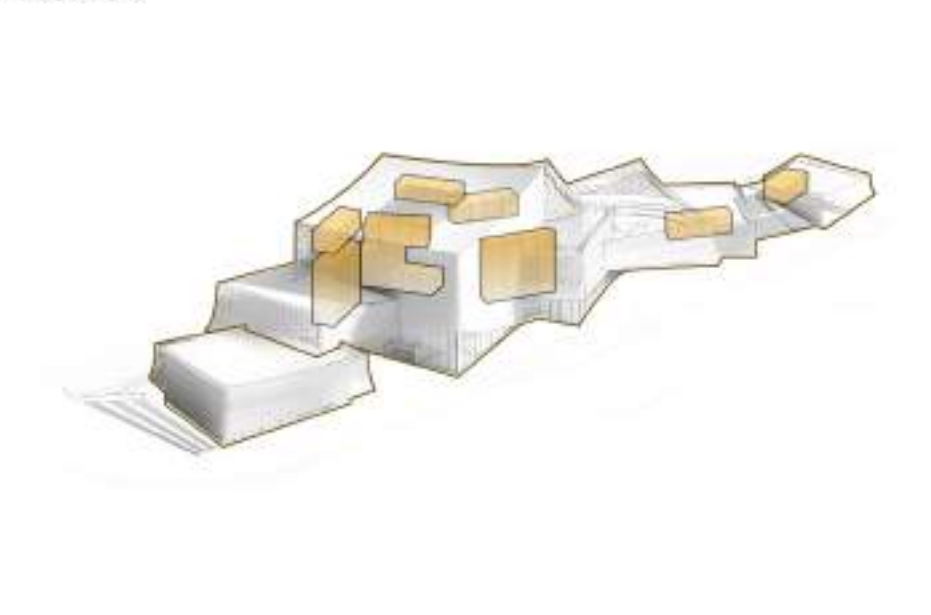
生态轻盈的双层玻璃幕墙



幕墙之间围合建筑的景观坡道



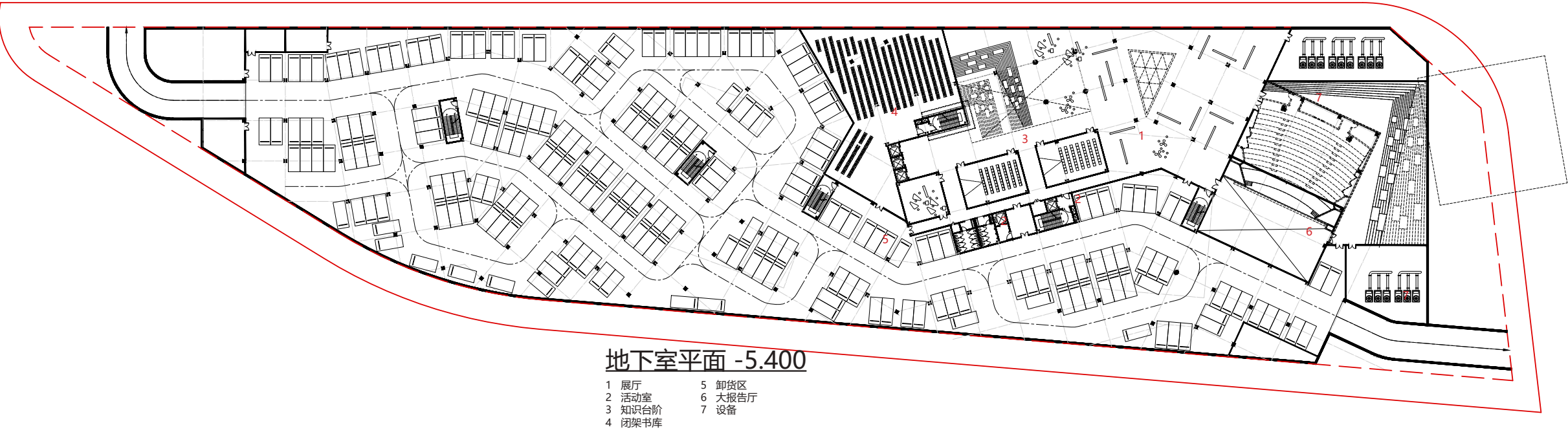
层叠的书架景观



通达室内的屋顶生态花园

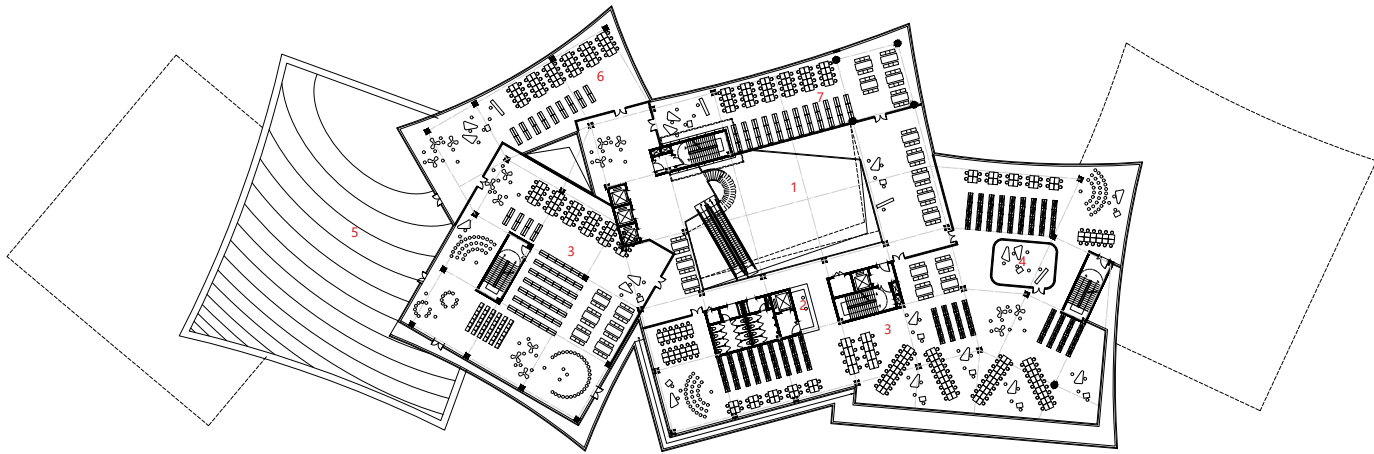






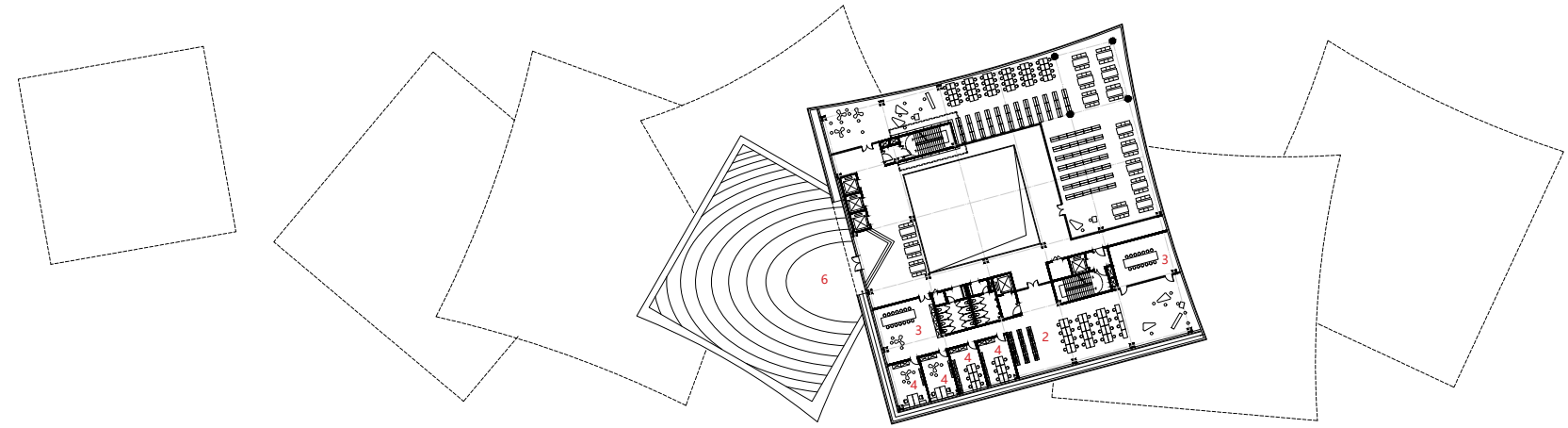
地下室平面 -5.400

- 1 展厅
- 2 活动区
- 3 科技展示区
- 4 闭路书库
- 5 时区
- 6 大报告厅
- 7 设备



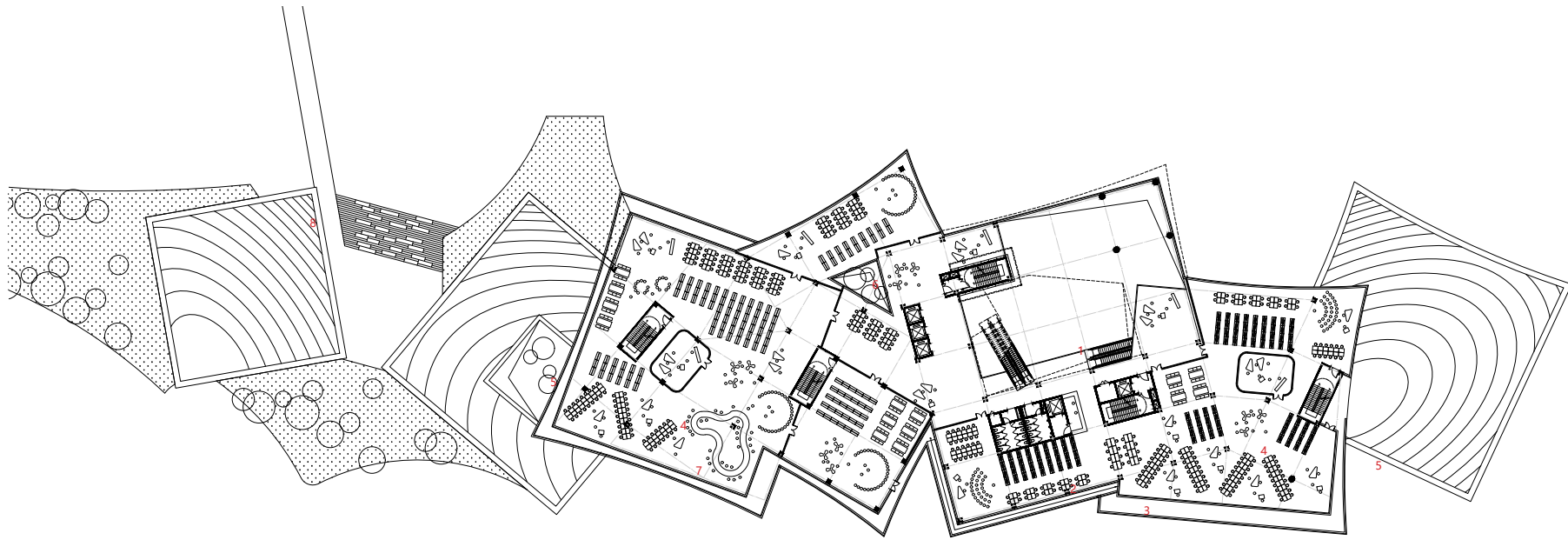
三层平面 +15.000

- 1 共享中庭
- 2 服务台
- 3 科技展示区
- 4 活动室
- 5 展览花园
- 6 有声图书馆
- 7 普通阅览室



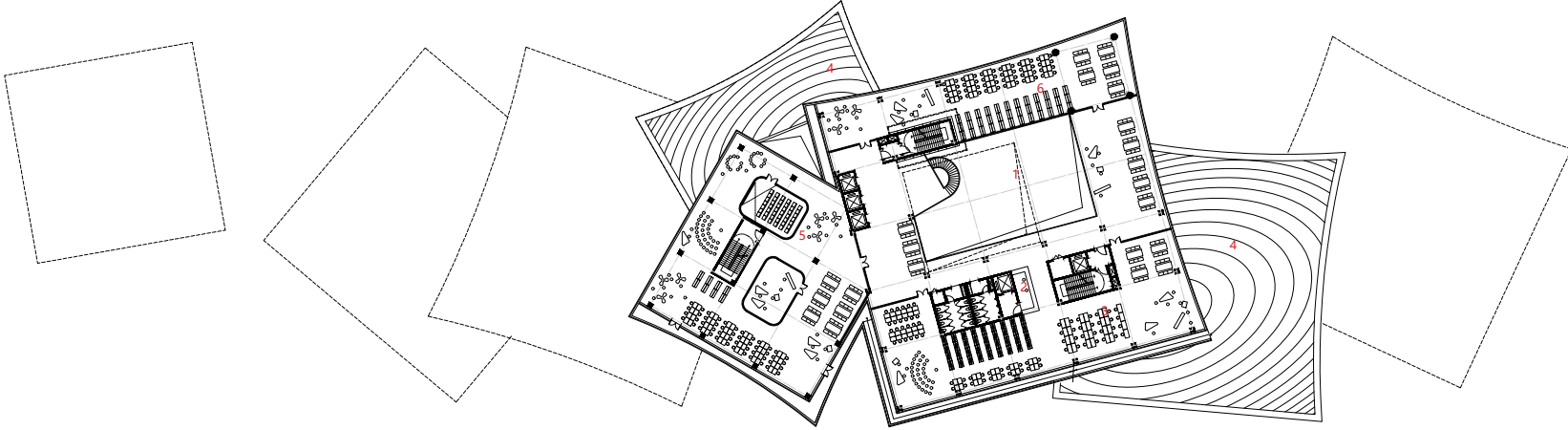
五层平面 +25.200

- 1 共享中庭
- 2 业务办公室
- 3 会议室
- 4 行政办公室
- 5 文藏馆
- 6 展览花园



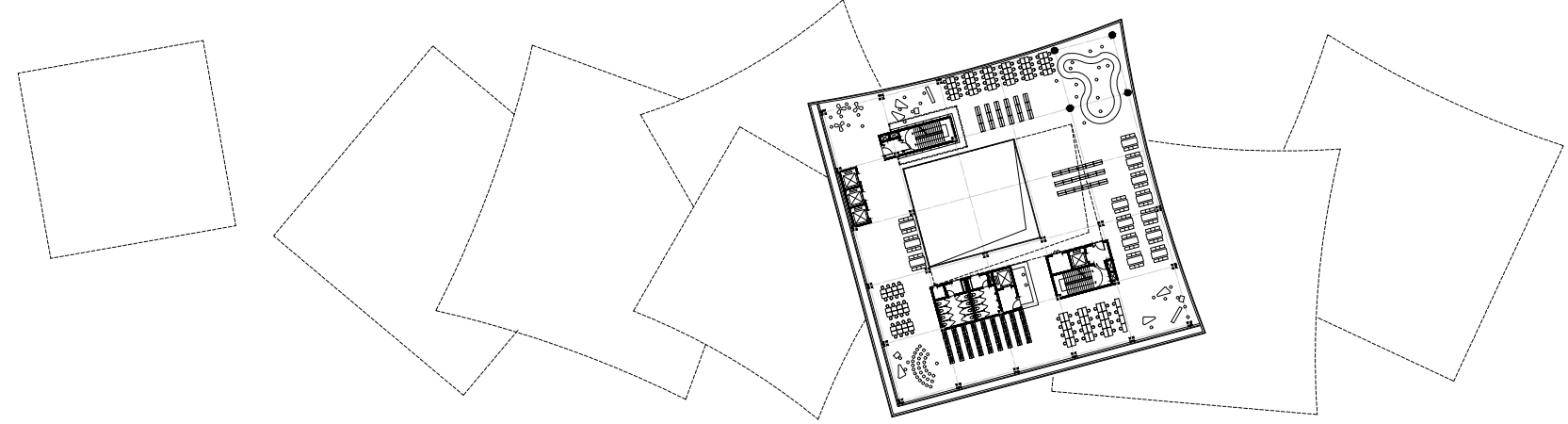
二层平面 +10.200

- 1 共享中庭
- 2 服务台
- 3 文史展示区
- 4 活动室
- 5 展览花园
- 6 有声图书馆
- 7 普通阅览室
- 8 过路天桥



四层平面 +20.400

- 1 共享中庭
- 2 服务台
- 3 科技展示区
- 4 展览花园
- 5 守山阁主展馆
- 6 特种阅览室



六层平面 +30.000

- 1 共享中庭
- 2 展览书吧

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